

WDW-2

Groupoid-Faithful Proper-Redescription Descent

Spatial reduction, certified internal-gauge extension, and boundary-standing non-collapse

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Exact scope. The theorem carries the exact WDW-1 source-attached canonical realization through proper spatial redescription. It proves regular-chart characteristic reduction and groupoid-faithful global descent while retaining isotropy, boundary charge, large transformations, and sector data. It does not claim normal-constraint closure, refoliation, quantization, or a physical Hilbert space.

Abstract

This paper constructs the proper-redescription quotient of the canonical Einstein–source realization established in WDW-1. The input is the exact branch-typed WDW-1 handoff, not an unrelated ADM surrogate: it retains the smooth canonical carrier, constraint candidates, connection and reconstruction ancestry, boundary ledger, and, for canonical matter, the map from complete histories into the structured Module-A source quotient. The unreduced canonical carrier is the regular cotangent bundle of the smooth tame configuration manifold. It carries the actual smooth identity-component diffeomorphism action, its cotangent lift, and an equivariant regular-dual moment map. The declared left action has moment map $J_{\text{left}} = -\mathcal{D}$, where $\mathcal{D}$ is the conventional positive-Lie-drag generator. A compatible Sobolev tower is used for tame elliptic estimates, Ebin slices, and strong Hilbert completions of the reduced slice charts; no differentiable action on one fixed full T^*H^s carrier is asserted. Direct variation recovers the WDW-1 momentum-constraint density and fixes the generator, moment-map, and boundary signs separately. On the free regular orbit-type locus, the elliptic orbit operator, its tame Green family, and slice-adapted cotangent coordinates give

$$\text{im } \rho_{\text{prop}} = \ker \omega_{\mathcal{C}_{\text{si}}} = \ker d\pi_{\text{prop}},$$

with a weakly symplectic smooth reduced form and a compatible strong Hilbert form on every completed Sobolev slice chart.

The global descent object is not presumed to be a Hausdorff manifold. We construct a boundary-fixed local-presentation groupoid, its explicit action on the spatial/internal zero locus, and the resulting phase/action groupoid. We prove the cover augmentation, action anchor, source, target, composition, orbit, tangent, and invariant-map universal properties before constructing the realization functor into the Gauge Transition Groupoid Theorem. Local slices remain computational presentations; isotropy and rank-changing strata remain in the action groupoid. Transformations with nonzero surface charge, boundary flux, mapping-class action, or registered sector change remain outside the proper groupoid. For complete canonical histories, proper foliation-preserving arrows leave the WDW-1 structured source class, attachment, witness, reconstruction ancestry, and pointed-lift identity unchanged. Hence the source-realization family descends fibrewise, including $\overline{\text{Src}}_{\phi}([\gamma_{\phi}]) = \hat{x}_{\phi}$.

The vacuum and attached Klein–Gordon branches occupy the spatial theorem with trivial internal factor. A nontrivial internal extension is admitted only through a populated certificate resolving its complete Dirac ledger, global group form, lift convention, boundary generator, and regularity data. The normal Hamiltonian-constraint family is equivariant under the completed spatial/internal quotient and is passed to WDW-3 with its lapse smearing transformed. Its self-bracket, full normal characteristic status, refoliation, quantum constraint operator, anomaly problem, physical Hilbert space, and clock interpretation are not claimed.

Keywords: AASC; canonical gravity; ADM; momentum map; symplectic reduction; diffeomorphism group; Lie groupoid; boundary charge; source realization; Wheeler–DeWitt.

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Read this first: exact endpoint

WDW-2 endpoint. The accepted WDW-1 vacuum and attached Klein–Gordon canonical realizations admit a source-realization-family-preserving, groupoid-faithful quotient by proper spatial redescription. On each populated free regular chart the proper orbit, characteristic, and quotient-vertical distributions coincide. Nonzero boundary, large, and sector data are not quotiented. The same theorem extends to an internal-gauge factor only when its independent certificate is populated.

Established in WDW-2	Not established in WDW-2
Smooth tame regular-cotangent carriers with compatible Sobolev towers	A differentiable full action on one fixed T^*H^s carrier
Spatial cotangent-lift moment map and proper-generator closure	Hamiltonian–Hamiltonian closure or full refoliation
Regular-chart characteristic equality with strong completed slice charts	A globally smooth or Hausdorff superspace
Explicit phase/action-groupoid descent without a global selector	Automatic quotienting of boundary, large, or sector-changing transformations
History-level descent of the WDW-1 source-realization family	A map from the entire model phase space to one fixed source class
Conditional theorem for independently certified internal sectors	Occupied Standard-Model or other nontrivial internal matter in this paper
Equivariant normal-constraint family handed to WDW-3	A Wheeler–DeWitt operator, regulator, anomaly theorem, or physical inner product

Part I

Endpoint, imports, and frozen comparison class

1. The theorem act and novelty boundary

Definition 1.1 (WDW-2 theorem act). The WDW-2 theorem act is the construction, on the exact eligible WDW-1 handoff, of:

1. a controlled field phase-space completion and common smooth functional domain;
2. the spatial and independently certified internal Hamiltonian actions;
3. the proper spatial/internal zero locus and its regular characteristic quotient;
4. a boundary-fixed proper-redescription groupoid retaining singular and standing-bearing data;
5. complete-history source and reconstruction descent; and
6. an equivariant normal-constraint family with the exact WDW-3 burden recorded.

The theorem is not a claim of historical priority for cotangent-lift reduction, the Ebin slice, the ADM momentum constraint, or Lie groupoids [1–3, 6, 8]. Its new content is the exact interface: one source-fibre-preserving WDW-1 object is carried through a field-theoretically typed local reduction, a global no-selector groupoid, a boundary/large-transformation guard, and a historywise source quotient without opening the normal-deformation theorem.

Definition 1.2 (Correctness-relevant criticism). A criticism is correctness-relevant when it preserves the frozen WDW-2 target and identifies a failed WDW-1 import, field completion, action, moment-map calculation, properness profile, regularity hypothesis, characteristic equality, groupoid bridge, source-history descent, boundary classification, or reportability claim.

Proposition 1.3 (Proof-carrier exhaustiveness). *Every correctness-relevant attack on the endpoint reaches one of four proof carriers: the imported WDW-1 object, conventional canonical field theory, the imported Gauge Transition Groupoid theorem, or a new WDW-2 interface certificate.*

Proof. The endpoint is the conjunction of exactly those four carriers. The imported WDW-1 object fixes the branch and ancestry. Canonical field theory supplies the action, moment map, regularity, and reduction. The transition theorem supplies the role-level local/restriction grammar. WDW-2 proves that the canonical object instantiates that grammar and preserves source and boundary standing. If the target is unchanged and the conjunction is false, at least one conjunct or one interface implication fails. A change of topology, boundary class, regularity scale, source branch, or quantum target changes the antecedent rather than refuting it. $\square$

2. How to read the proof

The paper is read in its dependency order, not from a familiar downstream formula backward. Sections 3–4 fix the exact WDW-1 object and its branch types. Sections 6–10 choose the analytic carriers on which a field-theoretic reduction theorem is meaningful. Sections 11–16 calculate the generators and determine which transformations are proper. Sections 17–22 prove the local characteristic quotient. Sections 23–27 construct the global groupoid and instantiate the imported role theorem. Sections 28–35 preserve source and reconstruction ancestry and form the WDW-3 handoff. Sections 36–39 state the adversarial and publication contracts. The appendices expose every calculation used by that chain.

Four distinctions govern the reading:

1. The smooth regular cotangent carrier supports the unreduced action, moment map, and reduction. The Sobolev tower supplies estimates and completed slice charts; it is not a fixed-index differentiable action carrier. The exact WDW-1 smooth domain is retained for curvature and normal-constraint formulas.
2. The occupied vacuum and Klein–Gordon branches have a trivial internal factor; the nontrivial internal theorem is certificate-gated and not reported as occupied.
3. Local regular quotients are smooth tame manifolds with compatible Hilbert slice completions; the global descent is an action groupoid that retains singular objects and isotropy.
4. The source map lives on complete histories; it is never extended to an assertion that one fixed source class is the whole model phase space.

An objection must be traced to the exact theorem carrier it challenges. The Gauge Transition Groupoid import cannot repair a failed moment-map calculation, and the local slice theorem cannot authorize source descent. Conversely, a request for the Hamiltonian self-bracket is not an objection to W2.4; it is the declared WDW-3 burden.

3. Dependency lock and exact nonclaims

The no-backflow order is

AASC kernel and realized physical interior $\implies$ accepted Einstein–source realization,

$$\text{WDW-1 W1.1–W1.8} \implies \mathfrak{H}_{\text{W1}} \xrightarrow{\text{Elig}_{23}} \mathfrak{B}_{\text{W2}}^{\text{in}},$$

$$\mathfrak{C}_{\text{field}}^2 + \mathfrak{C}_{\text{sp}}^2 + \mathfrak{C}_{\text{int}}^2 \implies (\mathcal{P}^\dagger, \Omega^\dagger, J_{\text{si}}, \mathcal{C}_{\text{si}}),$$

$$\mathfrak{C}_{\text{prop}}^2 + \mathfrak{C}_{\text{reg}}^2 \implies \text{im } \rho_{\text{prop}} = \ker \omega_{\mathcal{C}_{\text{si}}} = \ker d\pi_{\text{prop}},$$

$$\mathfrak{C}_{\text{bridge}}^2 + \mathfrak{C}_{\text{srcdesc}}^2 \implies \text{groupoid and source-family descent},$$

$$\text{WDW-2 output} \implies \text{WDW-3 input}.$$

No arrow reverses. Reduction does not authorize the Einstein equation, source attachment, canonical lift, or AASC kernel. A later quantum construction cannot repair a failed classical gate.

Nonclaim 3.1 (Frozen exclusions). No theorem below asserts the Hamiltonian–Hamiltonian bracket, full hypersurface-deformation closure, refoliation equivalence, completeness of lapse directions in the characteristic kernel, a global smooth superspace, absence of all singular or Gribov strata, universal internal-gauge occupancy, a quantum anomaly result, BRST cohomology, a Wheeler–DeWitt operator, a physical inner product, a preferred representative, or a preferred clock.

4. Exact WDW-1 import and source lineage

The imported predecessor is the final WDW-1 publication edition together with the declared Gauge Transition Groupoid theorem. Their load-bearing uses and prohibited promotions are stated explicitly below.

Definition 4.1 (Eligible input family). Import the WDW-1 disjoint handoff

$$\mathfrak{H}_{\text{W1}} = \mathfrak{H}_{\text{W1}}^{\text{vac}} \amalg \coprod_{\hat{x} \in I_{\text{can}}} \coprod_{\lambda \in \mathfrak{L}^+(\hat{x})} \mathfrak{H}_{\text{W1}}^{\text{can}}(\lambda) \amalg \coprod_{\hat{x} \in I_{\text{forced}}} \mathfrak{H}_{\text{W1}}^{\text{forced}}(\hat{x}).$$

The WDW-2 input is the typed subfamily

$$\mathfrak{B}_{\text{W2}}^{\text{in}} = \{\mathfrak{H} \in \mathfrak{H}_{\text{W1}} : \text{Elig}_{23}(\mathfrak{H}) = \text{ELIGIBLE}\}.$$

Thus it contains the vacuum handoff and every pointed strong canonical lift. There is no inclusion map from the forced-source summand.

Proposition 4.2 (Import fidelity). *The WDW-2 input preserves, without compression, the WDW-1 phase carrier, two-form, constraint candidates, common domain, connection certificate, reconstruction map, boundary certificate, and source-realization coordinates of every eligible branch.*

Proof. This is a typed import rather than an existence paraphrase. For the vacuum branch we retain $(\hat{x}_0, \mathcal{P}_g^\infty, \Omega_g, \Phi_g, \mathcal{C}_{\text{vac}}, \mathcal{N}, \mathcal{N}_+, \mathcal{X}, \text{Recon}_{\hat{x}_0}, \text{ConnCert}, \text{BdyCert})$. For a pointed lift $\lambda = (\mathfrak{M}, \text{Src}_{\mathfrak{M}}, \gamma, c_\lambda)$, we additionally retain the complete-history solution family, the source map, the exact source fibre, the model phase space, its Dirac ledger, and the distinguished history. The dense comparison maps introduced below have these smooth objects as their domain; they do not replace or quotient their ancestry. $\square$

5. Exact source-import ledger

Table 1: Load-bearing import ledger.

Source	Exact WDW-2 use	Prohibited promotion
WDW-1 W1.1	accepted dependency order and connection occupancy	does not prove a group action or quotient
WDW-1 W1.2	Cauchy source projections and Ward ledger	does not make a source class a phase-space point
WDW-1 W1.3	geometric tangential constraint relation	does not prove that relation is a moment map
WDW-1 W1.4	smooth gravitational carrier, two-form, and multiplier status	does not prove a field-theoretic characteristic quotient
WDW-1 W1.5	source-fibre-preserving prequotient reconstruction	does not identify distinct spatial presentations
WDW-1 W1.6	pointed total canonical lifts and complete Dirac ledger	does not resolve an internal sector left unresolved
WDW-1 W1.7	compact boundary standing and extension grammar	does not calculate a WDW-2 surface charge
WDW-1 W1.8	disjoint handoff family and downstream eligibility	eligibility is not proof that G_2 closes
The Realized Physical Interior	regular characteristic/coisotropic role architecture	does not promote a finite or local role theorem to ADM fields
Gauge Transition Groupoid definitions	continuation site, typed arrows, preservation, restriction, and composition grammar	do not supply a Lie action, symplectic form, or momentum map
Gauge Transition Groupoid operation propositions	identity, inverse, partial composition, associativity, restriction, and coherence burdens	do not establish those clauses for canonical fields without the bridge
Gauge realization stability theorem	path-dependent role realization is an obstruction	does not forbid registered physical holonomy or curvature
Boundary-fixed groupoid-layer theorem	minimal global role descent and hidden-selector exclusion	does not imply a globally smooth orbit manifold
Exhaustion of weaker transition realizations	failure classification for incomplete transition architectures	does not classify dynamics, measurement, or coarse-graining
SM-2 and SYM-1	proper/improper internal transformations, Gauss and boundary grammar	do not produce a curved-space canonical matter occupant
SYM-0	distinction among constraint, presentation, action, consequence, and obstruction	does not prove a generator bracket
SYM-2	target-update guard for dynamical spacetime, tetrads, Lorentz structure, and boundaries	flat-space data cannot be silently transported
The Physical Clock as a Quotient	boundary/external-time guard and relational-use discipline	does not prove spatial reduction or refoliation

Source	Exact WDW-2 use	Prohibited promotion
Covariant Differentiation from the AASC Kernel	standing-preserving transport and Levi–Civita ancestry	does not define a superspace connection or functional derivative

Every load-bearing excerpt is identified by source title, label, hypothesis summary, local instantiation, conclusion used, and prohibited stronger reading. The imports are not treated symmetrically. WDW-1 fixes the physical and canonical input; conventional field theory proves the new Hamiltonian and regularity statements; the Gauge Transition Groupoid theorem constrains the global descent architecture; the remaining corpus sources police role and target typing. This division prevents an AASC role theorem from being presented as an analytic field calculation and prevents a standard slice theorem from being presented as source or standing closure.

6. Branch partition and frozen target

Definition 6.1 (WDW-2 branch partition). For $b \in \mathfrak{B}_{W2}^{\text{in}}$, set

$$\begin{aligned}
I_2^{\text{sp}} &= \{b : \mathfrak{C}_{\text{sp}}^2(b) \text{ is populated}\}, \\
I_2^{\text{triv}} &= \{b \in I_2^{\text{sp}} : \mathcal{K}_b = 0\}, \\
I_2^{\text{joint}} &= \{b \in I_2^{\text{sp}} : \mathfrak{C}_{\text{int}}^2(b) \text{ is populated}\}, \\
I_2^{\text{blocked}} &= \mathfrak{B}_{W2}^{\text{in}} \setminus (I_2^{\text{triv}} \amalg I_2^{\text{joint}}).
\end{aligned}$$

The two passing classes are disjoint by the tag “trivial internal carrier” versus “populated nontrivial internal certificate.” An unresolved internal generator candidate is not set to zero; it is blocked.

Proposition 6.2 (Occupied branch status). *The WDW-1 vacuum handoff and the minimally coupled Klein–Gordon source-realization family belong to I_2^{triv} . No nontrivial internal branch is claimed occupied in this paper.*

Proof. WDW-1 gives the vacuum branch no internal carrier. Its scalar Dirac ledger proves that the Klein–Gordon model has no scalar primary, Gauss, second-class, or reducibility constraint; hence $\mathcal{K}_{\mathfrak{M}_\phi} = 0$. Sections 7–13 populate their common spatial certificate. The manuscript contains a conditional nontrivial-internal theorem but no model-specific internal occupation claim. $\square$

Definition 6.3 (Frozen WDW-2 target). For a passing branch b , the target object is

$$\begin{aligned}
\mathcal{T}_{W2}(b) = (b, \mathcal{P}_b^\dagger, \Omega_b^\dagger, \{\mathcal{Q}_b^r, S_{b,x}^r, T^* S_{b,x}^r\}_{r \geq r_0}, \mathcal{D}_b, \mathbb{A}_{\text{sp},b}, \mathbb{A}_{\text{int},b}, \mathbb{G}_{\text{pres}b}, \mathbb{A}_{\text{prop}b}, J_{\text{si},b}, \mathcal{C}_{\text{si}b}, \mathcal{C}_{\text{si}b}^{\text{reg}}, \pi_{\text{prop},b}, \\
\mathfrak{B}_b, \mathfrak{Q}_b, \mathfrak{S}_b, \mathcal{O}_{\text{prop},b}, \mathfrak{H}_{3,b}).
\end{aligned}$$

The compact occupied core has a connected compact oriented Cauchy three-manifold Σ without boundary, the WDW-1 sign and density conventions, the WDW-1 Levi–Civita ancestry, and either the vacuum or attached Klein–Gordon canonical branch.

Part II

Field carriers and regular action domains

7. Smooth tame carrier and Sobolev tower

Set

$$\mathcal{Q}_g^\infty = \text{Met}^\infty(\Sigma), \quad \mathcal{Q}_\phi^\infty = \text{Met}^\infty(\Sigma) \times C^\infty(\Sigma).$$

These are smooth tame Fréchet manifolds. If E_b is the configuration bundle, write $E_b^\vee \otimes \text{Dens}$ for its density-valued fibre dual.

Definition 7.1 (Regular cotangent phase carriers). For $b = g, \phi$, define

$$\mathcal{P}_b^\dagger = T^\dagger \mathcal{Q}_b^\infty = \mathcal{Q}_b^\infty \times \Gamma^\infty(E_b^\vee \otimes \text{Dens}).$$

This is the regular cotangent bundle, not the full continuous-dual cotangent bundle of a Fréchet manifold. Its canonical one-form and two-form are

$$\begin{aligned} \Theta_b^\dagger(\delta q, \delta p) &= \langle p, \delta q \rangle, & \Omega_b^\dagger &= d\Theta_b^\dagger, \\ \Omega_b^\dagger((u, \alpha), (v, \beta)) &= \langle \alpha, v \rangle - \langle \beta, u \rangle. \end{aligned}$$

The density pairing is intrinsic. The form is smooth tame and weakly symplectic: its musical map is injective on the regular tangent carrier, but no surjection onto the full continuous dual is claimed.

Definition 7.2 (Compatible Sobolev tower). Fix $r_0 > 7/2$ and set, for every integer $r \geq r_0$,

$$\mathcal{Q}_g^r = \text{Met}^r(\Sigma), \quad \mathcal{Q}_\phi^r = \text{Met}^r(\Sigma) \times H^r(\Sigma).$$

Then

$$\mathcal{Q}_b^\infty = \varinjlim_{r \geq r_0} \mathcal{Q}_b^r, \quad \text{Diff}_0^\infty(\Sigma) = \varinjlim_{r \geq r_0} \text{Diff}_0^{r+1}(\Sigma).$$

The finite levels are used only for tame estimates, elliptic closed-range and splitting estimates, Ebin slice completions, and the strong cotangent completions $T^*S_{b,x}^r$ of regular reduced slice charts. They are not asserted to carry a differentiable action on the full unreduced $T^*\mathcal{Q}_b^r$.

Remark 7.3 (Sign convention). WDW-1 uses $\Omega = \int \delta\pi \wedge \delta h$ and $\iota_{X_F}\Omega = -dF$. We retain it. For the left action $\varphi \cdot q = (\varphi^{-1})^*q$, the conventional generator $\mathcal{D}[\xi] = \langle p, \mathcal{L}_\xi q \rangle$ satisfies $X_{\mathcal{D}[\xi]} = -\rho(\xi)$. Hence the left-action moment map is

$$\langle J_{\text{left}}, \xi \rangle = -\mathcal{D}[\xi], \quad \iota_{\rho(\xi)}\Omega = -d\langle J_{\text{left}}, \xi \rangle = d\mathcal{D}[\xi].$$

This convention is used everywhere below.

Definition 7.4 (WDW-1 identification and comparison maps). Let $\mathcal{P}_g^\infty$ and $\mathcal{P}_\phi^\infty$ be the exact smooth WDW-1 carriers. Their field and momentum bundles identify canonically with $\mathcal{P}_g^\dagger$ and $\mathcal{P}_\phi^\dagger$; denote those ancestry-preserving identifications by $\iota_b^\dagger$. For every $r \geq r_0$, configuration restriction gives a continuous dense comparison $\iota_{b,Q}^r : \mathcal{Q}_b^\infty \rightarrow \mathcal{Q}_b^r$, and smooth density momenta define continuous functionals on the relevant completed slice tangents. Let $\mathcal{D}_b$ be the exact WDW-1 common functional domain, enlarged only by smooth compact-core smearings.

Proposition 7.5 (Carrier, tower, and comparison). *Each $(\mathcal{P}_b^\dagger, \Omega_b^\dagger)$ is a weakly symplectic smooth tame Fréchet manifold and is the exact regular-cotangent realization of the corresponding smooth WDW-1 carrier. The configuration maps $\iota_{b,Q}^r$ have dense image and the Sobolev tower is compatible under restriction. On every regular slice, the smooth regular cotangent form extends to the canonical strong symplectic form on $T^*S_{b,x}^r$. No comparison changes reconstruction or source ancestry.*

Proof. $\text{Met}^\infty(\Sigma)$ is an open subset of the tame Fréchet space of smooth symmetric two-tensors; the scalar factor is tame. Smooth density pairing makes $\Omega_b^\dagger$ injective on the regular tangent carrier. It is not promoted to a strong Fréchet form. Smooth sections are dense in each finite Sobolev level, positivity is open, and the restriction maps commute. The elliptic slice construction in Appendix G supplies compatible smooth slices $S_{b,x}^\infty$ and Hilbert completions $S_{b,x}^r$. The canonical form on the regular slice cotangent is the restriction of the strong canonical form on $T^*S_{b,x}^r$. Finally, $\iota_b^\dagger$ changes no structured source, attachment, witness, connection, or reconstruction coordinate; those remain attached to $\mathcal{D}_b$ and complete histories. $\square$

Certificate 7.6 (Populated field certificate). $\mathfrak{C}_{\text{field}}^2(b)$ contains $\mathcal{P}_b^\dagger, \Omega_b^\dagger, \{\mathcal{Q}_b^r\}_{r \geq r_0}, \iota_b^\dagger, \mathcal{D}_b$, weak nondegeneracy, slice-completion compatibility, the bracket domain, and reconstruction compatibility. Proposition 7.5 populates it for $b = g, \phi$.

8. Common domain and distribution-valued constraints

Definition 8.1 (Common functional domain). $\mathcal{D}_b$ is the dense smooth submanifold on which the WDW-1 normal and spatial constraint densities, their first variations, all brackets used in this paper, and the complete-history reconstruction map are defined. The spatial moment map is a smooth tame regular-dual map on $\mathcal{P}_b^\dagger$. The normal family is used only on $\mathcal{D}_b$.

Proposition 8.2 (Derivative-loss accounting). *No differentiable spatial action on one fixed $T^*\mathcal{Q}_b^r$ and no normal Hamiltonian vector field on a full Sobolev cotangent completion are required by WDW-2. Instead,*

$$(q, \xi) \in \mathcal{Q}_b^{r+1} \times H^{r+1}(T\Sigma) \mapsto \mathcal{L}_\xi q \in H^r$$

is recorded as a one-level-loss estimate. The spatial action and moment map are differentiated only on $\mathcal{P}_b^\dagger$; Sobolev levels are used after elliptic splitting to complete the reduced slices. Every bracket involving $\mathcal{H}_\perp$ remains on $\mathcal{D}_b$.

Proof. For generic $q \in H^r$, $\mathcal{L}_\xi q$ has only H^{r-1} regularity; increasing one fixed index does not turn the full pullback action into a smooth Banach action [5]. On smooth fields, pullback and Lie differentiation are smooth tame and remain in the smooth tangent bundle. Elliptic estimates for the orbit operator are compatible across the Sobolev tower, so their inverse-limit splitting and completed slice restrictions are legitimate. The scalar-curvature and quadratic-momentum terms of $\mathcal{H}_\perp$ still require the exact WDW-1 smooth domain. Thus spatial transport loss and normal-functional regularity are separately exposed. $\square$

9. Spatial transformation group

Let

$$\mathcal{G}_{\text{sp}}^\infty = \text{Diff}_0^\infty(\Sigma).$$

We use the identity component because mapping-class transformations are standing-bearing large data unless separately certified otherwise.

Definition 9.1 (Spatial action). On configurations,

$$\varphi \cdot h = (\varphi^{-1})^* h, \quad \varphi \cdot \phi = (\varphi^{-1})^* \phi.$$

The action on momenta is the regular cotangent lift. Its infinitesimal configuration action for $\xi \in \Gamma^\infty(T\Sigma)$ is

$$\rho_{\mathcal{Q}}(\xi)q = -\mathcal{L}_\xi q.$$

On phase space,

$$\rho_{\text{sp},b}(\xi)(q, p) = (-\mathcal{L}_\xi q, -\mathcal{L}_\xi p).$$

The minus signs are forced by the chosen left action. The Hamiltonian generator $\mathcal{D}_b[\xi]$ displayed in Section 12 produces the conventional positive Lie drag, so

$$X_{\mathcal{D}_b[\xi]} = -\rho_{\text{sp},b}(\xi)$$

under the WDW-1 Hamiltonian convention.

Proposition 9.2 (Smooth tame cotangent action). *The pullback action of $\text{Diff}_0^\infty(\Sigma)$ on $\mathcal{Q}_b^\infty$ is smooth tame. Its lift to $\mathcal{P}_b^\dagger$ is the ordinary tensor-density pullback, preserves $\Theta_b^\dagger$ and $\Omega_b^\dagger$, and has $\rho_{\mathcal{Q}}(\xi)q = -\mathcal{L}_\xi q$ as a genuine smooth tangent vector.*

Proof. Composition, inversion, tensor pullback, and differentiation are smooth tame on the inverse-limit group and smooth section spaces [3, 4]. Lie differentiation therefore remains in $T_q\mathcal{Q}_b^\infty$. Naturality of the density pairing gives

$$\langle (\varphi^{-1})^* p, (\varphi^{-1})^* \delta q \rangle = \langle p, \delta q \rangle,$$

so the regular cotangent lift preserves $\Theta_b^\dagger$ and its exterior derivative. Proposition 8.2 records why this proof is not transferred to one fixed Sobolev level. $\square$

10. Internal automorphism groupoid

Definition 10.1 (Trivial occupied internal factor). For the occupied vacuum and Klein–Gordon branches,

$$\mathbb{A}_{\text{int},b} = 1, \quad \mathfrak{g}_{\text{int},b} = 0, \quad J_{\text{int},b} = 0.$$

This is an occupied trivial certificate, not evidence for a nontrivial Gauss sector.

Definition 10.2 (Conditional internal certificate). For a nontrivial canonical WDW-1 branch, $\mathfrak{C}_{\text{int}}^2(b)$ consists of the actual bundle automorphism groupoid, its global form and action on every canonical variable, a differentiable generator $\mathcal{G}_b[\alpha]$, a moment map $J_{\text{int},b}$, the completely resolved internal Dirac ledger, reducibility data, lift convention, common bracket domain, boundary improvement, charge, large-gauge components, and sector action.

Proposition 10.3 (Internal stop rule). *If any coordinate of Definition 10.2 is unresolved, the branch lies in I_2^{blocked} and contributes no internal quotient theorem.*

Proof. An unresolved second-class direction is not a characteristic gauge direction; an unresolved global form does not determine the large-gauge components; and a nondifferentiable generator does not define a Hamiltonian action. Each missing coordinate can therefore change the zero locus, orbit distribution, boundary standing, or quotient. It is tensor rather than skin and cannot be filled by notation. $\square$

11. Orbit types, slices, and the regular locus

Definition 11.1 (Free tame-regular locus). Let $\mathcal{Q}_{b,\text{fr}}^\infty$ be the open subset on which the stabilizer in $\text{Diff}_0^\infty(\Sigma)$ is trivial and the infinitesimal orbit operator R_q has the elliptic tame splitting of Appendix G. Let

$$\mathcal{P}_{b,\text{fr}}^\dagger = T^\dagger \mathcal{Q}_{b,\text{fr}}^\infty, \quad \mathcal{C}_{\text{si}b}^{\text{reg}} = J_{\text{sp},b}^{-1}(0) \cap \mathcal{P}_{b,\text{fr}}^\dagger.$$

Other constant orbit-type strata may be treated by the same method with isotropy retained, but the occupied equality below is stated on the free stratum.

Proposition 11.2 (Nonempty occupied regular locus). *For both $b = g$ and $b = \phi$, the free regular zero locus is nonempty.*

Proof. Smooth metrics with trivial isometry group form an open dense subset of the metric space in the relevant regularity regime [2]. Choose such an h , take $\pi = 0$, and, in the scalar model, take $p = 0$ with any scalar whose joint stabilizer remains trivial (in particular the metric already removes identity-component stabilizers). The regular cotangent momentum map vanishes at the zero covector. The Ebin slice and elliptic Green operator supply the tame split orbit and its compatible Hilbert slice completions. This populates a WDW-2 spatial zero-locus chart; it does not assert that the selected point also solves the still-downstream normal constraint. $\square$

Remark 11.3 (Occupied solution histories). Complete vacuum and Klein–Gordon histories supplied by WDW-1 enter the history groupoid wherever their evaluations meet a declared regular chart. The regular reduction theorem is chartwise and does not require the distinguished history γ_ϕ to have trivial isotropy at every time.

Part III

Generators, moment maps, and proper closure

12. Direct spatial-generator calculation

For the gravitational and scalar configurations define, on the smooth common domain,

$$\mathcal{D}_g[\xi] = \int_{\Sigma} \pi^{ij} \mathcal{L}_{\xi} h_{ij}, \quad (1)$$

$$\mathcal{D}_{\phi}[\xi] = \int_{\Sigma} \left(\pi^{ij} \mathcal{L}_{\xi} h_{ij} + p_{\phi} \mathcal{L}_{\xi} \phi \right). \quad (2)$$

These formulas are the cotangent-lift Hamiltonians. They are not inferred from the geometric source relation.

Lemma 12.1 (Gravitational integration by parts). *On the compact boundaryless core,*

$$\mathcal{D}_g[\xi] = \int_{\Sigma} \xi^i (-2D_j \pi^j_i) = \int_{\Sigma} \xi^i \mathcal{H}_i^g.$$

Proof. Because π^{ij} is symmetric and has density weight one,

$$\pi^{ij} \mathcal{L}_{\xi} h_{ij} = \pi^{ij} (D_i \xi_j + D_j \xi_i) = 2\pi^{ij} D_i \xi_j.$$

The density divergence theorem gives

$$2\pi^{ij} D_i \xi_j = 2D_i (\pi^{ij} \xi_j) - 2\xi_j D_i \pi^{ij}.$$

The integral of the first term vanishes when $\partial\Sigma = \emptyset$. Relabeling the free index and lowering with h yields the displayed WDW-1 density $\mathcal{H}_i^g = -2D_j \pi^j_i$. $\square$

Lemma 12.2 (Scalar integration by parts and source match). *For the minimally coupled scalar branch,*

$$\mathcal{D}_{\phi}[\xi] = \int_{\Sigma} \xi^i (-2D_j \pi^j_i + p_{\phi} D_i \phi).$$

On every complete WDW-1 Klein–Gordon history,

$$\mathcal{H}_i^{\phi} = p_{\phi} D_i \phi = -\sqrt{h} j_{i,\phi}.$$

Proof. $\mathcal{L}_{\xi} \phi = \xi^i D_i \phi$, so only the gravitational term requires integration by parts. The WDW-1 scalar Legendre formula is

$$p_{\phi} = \sqrt{h} n^{\mu} \nabla_{\mu} \phi.$$

For $T_{\mu\nu} = \nabla_{\mu} \phi \nabla_{\nu} \phi - \frac{1}{2} g_{\mu\nu} ((\nabla \phi)^2 + m^2 \phi^2)$, the source projection convention gives

$$j_i = -h_i^{\mu} n^{\nu} T_{\mu\nu} = -(D_i \phi) (n^{\nu} \nabla_{\nu} \phi).$$

Multiplication by $\sqrt{h}$ proves the second identity. Thus the canonical scalar term matches the exact Module-A source projection on histories, while its existence still comes from the scalar canonical lift rather than from stress energy alone. $\square$

Proposition 12.3 (Boundary completion). *If $\partial\Sigma \neq \emptyset$, the variation of the bulk expression produces*

$$\delta \mathcal{D}_b^{\text{bulk}}[\xi] = \iota_{\rho_b(\xi)} \Omega_b + \delta B_{\partial,b}^{\text{diff}}[\xi].$$

A differentiable generator exists only after a branch-specific $Q_{\partial,b}^{\text{diff}}[\xi]$ satisfies $\delta Q_{\partial,b}^{\text{diff}} = -\delta B_{\partial,b}^{\text{diff}}$.

Proof. The integration by parts above and the corresponding variation retain a spatial surface term whenever the boundary is nonempty. The Regge–Teitelboim differentiability criterion requires that term to be an exact variation on the chosen boundary phase space [9]. Without the improvement, the bulk formula has no Hamiltonian vector field and cannot enter the proper algebra. The compact occupied core has $B_{\partial} = Q_{\partial} = 0$. For the improved conventional generator $\mathcal{D}_b^{\text{imp}} = \mathcal{D}_b^{\text{bulk}} + Q_{\partial,b}^{\text{diff}}$, the displayed convention gives $\delta \mathcal{D}_b^{\text{imp}} = \iota_{\rho_b} \Omega_b$, and the corresponding left-action moment map is $-\mathcal{D}_b^{\text{imp}}$. $\square$

13. Spatial moment-map theorem

Definition 13.1 (Spatial moment map). For $z = (q, p) \in \mathcal{P}_b^\dagger$ and $\xi \in \Gamma^\infty(T\Sigma)$, define

$$\langle J_{\text{left},b}(z), \xi \rangle = -\langle p, \mathcal{L}_\xi q \rangle = -\mathcal{D}_b[\xi].$$

We write $J_{\text{sp},b} = J_{\text{left},b}$ throughout the reduction. Its codomain is the regular dual

$$\mathfrak{X}^\infty(\Sigma)^\dagger = \Gamma^\infty(T^*\Sigma \otimes \text{Dens}),$$

paired with smooth vector fields. The WDW-1 momentum-constraint density $\mathcal{H}_{i,b}^{\text{tot}}$ represents the conventional generator $\mathcal{D}_b$; its negative represents $J_{\text{left},b}$.

Proposition 13.2 (Cotangent moment-map identity). *The map $J_{\text{left},b}$ is smooth tame and equivariant. With the frozen WDW-1 sign convention and the left action of Definition 9.1,*

$$\iota_{\rho_{\text{sp},b}(\xi)}\Omega_b^\dagger = -d\langle J_{\text{left},b}, \xi \rangle = d\mathcal{D}_b[\xi].$$

On the smooth domain the conventional generator density and the left-action moment-map density are, respectively,

$$\mathcal{H}_{i,b}^{\text{tot}} = -2D_j\pi^j_i + \begin{cases} 0, & b = g, \\ p_\phi D_i\phi, & b = \phi. \end{cases}, \quad \mathcal{J}_{i,b}^{\text{left}} = -\mathcal{H}_{i,b}^{\text{tot}}.$$

Proof. The regular cotangent lift preserves $\Theta_b^\dagger$. For the infinitesimal lifted action, Cartan's identity gives

$$0 = \mathcal{L}_{\rho(\xi)}\Theta_b^\dagger = \iota_{\rho(\xi)}d\Theta_b^\dagger + d(\iota_{\rho(\xi)}\Theta_b^\dagger).$$

Because the declared left action has configuration infinitesimal $-\mathcal{L}_\xi q$, while the conventional generator is $\mathcal{D}_b[\xi] = \langle p, \mathcal{L}_\xi q \rangle$, one has

$$\iota_{\rho(\xi)}\Theta_b^\dagger = -\mathcal{D}_b[\xi].$$

Cartan's identity therefore gives $\iota_{\rho(\xi)}\Omega_b^\dagger = d\mathcal{D}_b[\xi]$, which is $-d\langle J_{\text{left},b}, \xi \rangle$ exactly. Naturality of the cotangent construction gives

$$J_{\text{left},b}(\varphi \cdot z) = \text{Ad}_\varphi^* J_{\text{left},b}(z).$$

Lemmas 12.1 and 12.2 identify $\mathcal{H}_{i,b}^{\text{tot}}$, and negation identifies $\mathcal{J}_{i,b}^{\text{left}}$. No normal constraint or source-forcing equation occurs in this proof. $\square$

Certificate 13.3 (Populated spatial certificate). For $b = g, \phi$, $\mathfrak{C}_{\text{sp}}^2(b)$ consists of $\text{Diff}_0^\infty(\Sigma)$, the regular cotangent lift, the direct generator $\mathcal{D}_b[\xi]$, the left-action moment map $J_{\text{left},b} = -\mathcal{D}_b$, Proposition 13.2, equivariance, and the compact boundary ledger, together with the Sobolev loss and slice-completion ledgers. It is populated.

14. Ordinary and gauge-covariant diffeomorphisms

The occupied branches have no internal connection, so their spatial generator is unambiguous. For a conditional gauge branch the distinction is load-bearing.

Definition 14.1 (Lift certificate). If A is an internal connection and $\mathcal{G}[\alpha]$ its certified Gauss generator, define

$$\mathcal{D}^{\text{cov}}[\xi] = \mathcal{D}^{\text{ord}}[\xi] + \mathcal{G}[\iota_\xi A]$$

with the sign replaced if the branch convention uses the opposite infinitesimal internal action. $\mathfrak{C}_{\text{lift}}^2(b)$ records both actions, the lift of spatial diffeomorphisms to the bundle, the field dependence of $\iota_\xi A$, and the resulting joint bracket.

Proposition 14.2 (No double counting). *On an admitted internal branch, using $\mathcal{D}^{\text{cov}}$ and independently adding all Gauss directions defines the same joint orbit distribution as using $\mathcal{D}^{\text{ord}}$ and Gauss directions, but the generator brackets differ by the explicit field-dependent parameter term. A proof may use either convention, not both interchangeably.*

Proof. The displayed generators differ by a Gauss generator, so their spans after adjoining every Gauss direction coincide. However, the parameter $\iota_{\xi}A$ depends on the phase variable. Its Poisson bracket contributes the variation of that parameter and alters the structure function in the spatial-spatial bracket. Suppressing the term can misidentify the algebra even though the final distribution is the same. $\mathfrak{C}_{\text{lift}}^2$ therefore fixes one convention before the bracket calculation. $\square$

15. Internal generator calculation

Proposition 15.1 (Conditional internal cotangent theorem). *Let a WDW-1 branch carry a populated $\mathfrak{C}_{\text{int}}^2(b)$ after all matter-internal second-class constraints have been resolved. Then its cotangent-lift internal action has a differentiable equivariant moment map satisfying*

$$\iota_{\rho_{\text{int},b}(\alpha)}\Omega_b^\dagger = -d\langle J_{\text{int},b}, \alpha \rangle,$$

and its smooth generator is the improved Gauss functional recorded by the certificate.

Proof. On the resolved canonical carrier, the internal automorphism groupoid acts on the configuration variables by the formulas retained in $\mathfrak{C}_{\text{int}}^2(b)$. Its cotangent lift preserves the canonical one-form. The Cartan calculation of Proposition 13.2 therefore yields the signed moment-map identity. Direct integration by parts in the model-specific formula separates a bulk Gauss density from the surface variation; the certificate supplies the improvement and proves equivariance or records the exact central cocycle. Completeness of the Dirac ledger ensures that no second-class direction is included as an orbit direction. This theorem is conditional: there is no nontrivial internal occupant in the present scope. $\square$

16. Joint proper generator algebra

Definition 16.1 (Joint moment map). On I_2^{triv} , set $J_{\text{si},b} = J_{\text{sp},b}$. On a populated joint branch, set

$$\langle J_{\text{si},b}, (\xi, \alpha) \rangle = -\mathcal{D}_b[\xi] + \langle J_{\text{int},b}, \alpha \rangle,$$

using the actual bundle-automorphism extension and the convention fixed by $\mathfrak{C}_{\text{lift}}^2(b)$; no split semidirect product is presumed. The conventional generator algebra remains written in terms of $\mathcal{D}_b$ and the certified improved Gauss generator $\mathcal{G}_b$.

Proposition 16.2 (Spatial closure). *On the smooth common domain of the occupied branches,*

$$\{\mathcal{D}_b[\xi], \mathcal{D}_b[\eta]\} = \mathcal{D}_b[[\xi, \eta]]$$

in the WDW-1 Poisson convention. There is no central term on the compact core.

Proof. The Hamiltonian vector field of $\mathcal{D}_b[\xi]$ is the cotangent lift of Lie dragging by ξ . Commutators of lifted fundamental fields are the lift of $[\xi, \eta]$. Direct functional differentiation of (1)–(2) gives the same result. No surface cocycle exists because $\partial\Sigma = \emptyset$. $\square$

Proposition 16.3 (Conditional joint closure). *For a populated internal certificate, the proper generator family closes on the declared common domain with the exact extension bracket. In a split ordinary-diffeomorphism presentation the formulas are*

$$\begin{aligned} \{\mathcal{D}[\xi], \mathcal{D}[\eta]\} &= \mathcal{D}[[\xi, \eta]], \\ \{\mathcal{D}[\xi], \mathcal{G}[\alpha]\} &= \mathcal{G}[\mathcal{L}_\xi \alpha], \\ \{\mathcal{G}[\alpha], \mathcal{G}[\beta]\} &= \mathcal{G}[[\alpha, \beta]], \end{aligned}$$

modified by the named field-dependent and boundary terms when the extension is nonsplit.

Proof. The first line is Proposition 16.2. The second is naturality of the internal parameter under the certified spatial lift. The third is the closure coordinate of $\mathfrak{C}_{\text{int}}^2$. A nonsplit extension replaces the displayed direct formulas by its Atiyah or groupoid bracket; $\mathfrak{C}_{\text{lift}}^2$ records the correction. The branch passes only if the boundary cocycle vanishes identically on the proper subalgebra. Thus the statement does not infer closure from the word “gauge”; it packages the independently verified bracket. $\square$

17. Proper, improper, and large transformations

Definition 17.1 (Proper profile). For each admitted branch define

$$\mathfrak{g}_{\text{prop},b} = \left\{ \epsilon : \begin{array}{l} \epsilon \text{ preserves the frozen target and boundary conditions,} \\ Q_{\partial}[\epsilon] \equiv 0, \quad \delta Q_{\partial}[\epsilon] \equiv 0, \quad \text{Flux}_{\partial}[\epsilon] = 0, \\ \epsilon \text{ lies in the registered identity component and preserves sector data} \end{array} \right\}.$$

The complement in the differentiable transformation algebra is improper or standing-bearing; disconnected transformations are registered separately.

Proposition 17.2 (Compact-core properness). *For the occupied compact boundaryless core, every element of the connected spatial group $\text{Diff}_0^\infty(\Sigma)$ satisfying the branch-preservation conditions has $Q_{\partial} = \delta Q_{\partial} = \text{Flux}_{\partial} = 0$. Mapping-class transformations $\text{Diff}^\infty(\Sigma)/\text{Diff}_0^\infty(\Sigma)$ are not included in the proper groupoid.*

Proof. There is no spatial boundary on which a surface integral or flux can be supported. The connected action preserves orientation, topology, WDW-1 ancestry, and the registered source family by construction. A disconnected component can act on mapping-class or sector data even when the local constraint formula is unchanged. Since WDW-2 has no certificate that this action is skin, it is retained rather than quotiented. $\square$

Proposition 17.3 (Boundary non-collapse). *On any finite or asymptotic extension, a transformation with nonzero differentiable charge, nonzero charge variation, nonzero flux, boundary action, central term, or registered sector change is outside $\mathfrak{g}_{\text{prop},b}$.*

Proof. Each listed coordinate changes a differentiable generator, continuation law, or standing-bearing boundary/sector value. Quotienting it would identify states separated by the frozen invariant family. Definition 17.1 therefore excludes it. This proposition is a classification theorem; it does not claim that every boundary extension has been populated. Each such extension still requires its own BdyCert and explicit charge calculation. $\square$

Theorem W2. 1 (Eligible field-carrier and branch-occupation theorem). For every $b \in \mathfrak{B}_{\text{W2}}^{\text{in}}$ with a populated $\mathfrak{C}_{\text{field}}^2(b)$, the exact smooth WDW-1 carrier is identified with the weakly symplectic regular cotangent carrier $(\mathcal{P}_b^\dagger, \Omega_b^\dagger)$, equipped with a compatible Sobolev tower and strong Hilbert completions of every regular reduced slice chart, with its source and reconstruction ancestry unchanged. The branch partition

$$I_2^{\text{triv}} \amalg I_2^{\text{joint}} \amalg I_2^{\text{blocked}}$$

is disjoint and exhaustive. The vacuum and attached Klein–Gordon branches occupy I_2^{triv} and have nonempty free regular spatial zero-locus charts.

Proof. Proposition 7.5 gives the completion and comparison. Definition 6.1 and the complete WDW-1 internal-carrier ledger give the partition. Proposition 6.2 identifies the occupied branches, and Proposition 11.2 populates their regular charts. The forced WDW-1 branch has no inclusion into Definition 4.1. $\square$

Theorem W2. 2 (Spatial/internal Hamiltonian-action theorem). On the common domain, direct variation gives the conventional spatial generator $\mathcal{D}_b$; the spatial moment map of the declared left action is $J_{\text{left},b} = -\mathcal{D}_b$ and satisfies Proposition 13.2. For every populated nontrivial internal certificate, Proposition 15.1 supplies the internal moment map. Their joint map is typed by the actual bundle-automorphism extension and by one fixed ordinary/covariant lift convention. No source-forcing relation is used as a matter moment map.

Proof. Lemmas 12.1–12.2 provide the complete occupied variation. Proposition 13.2 proves Hamiltonianity and equivariance. Propositions 14.2 and 15.1 supply the conditional internal statement. The scalar term comes from its canonical cotangent carrier and is only then matched historywise to the Module-A source projection. $\square$

Theorem W2. 3 (Proper-generator closure and boundary-standing theorem). The occupied proper spatial generator family closes on $\mathcal{D}_g$ and $\mathcal{D}_\phi$. A nontrivial internal family closes only when its independent certificate proves the exact joint bracket. Nonzero boundary charge, flux, central term, mapping-class action, large internal action, or registered sector change excludes the transformation from the proper groupoid.

Proof. Propositions 16.2 and 16.3 give the algebra. Propositions 17.2 and 17.3 give the proper/improper and large-transformation classification. $\square$

Part IV

Regular characteristic reduction

18. Spatial/internal zero locus

Definition 18.1 (Partial constraint locus). For every passing branch,

$$\mathcal{C}_{\text{si}b} = J_{\text{si},b}^{-1}(0).$$

This is the spatial/internal zero locus. It is not the full WDW-1 preconstraint surface and does not impose or quotient the normal Hamiltonian constraint.

Proposition 18.2 (Relation to WDW-1 constraints). *On the smooth occupied domain,*

$$\mathcal{C}_{\text{pre},b} \subseteq \mathcal{C}_{\text{si}b} \cap \mathcal{D}_b.$$

The reverse inclusion is not asserted.

Proof. The WDW-1 preconstraint map contains both $\mathcal{H}_\perp^{\text{tot}}$ and $\mathcal{H}_i^{\text{tot}}$. Theorem W2.2 identifies the latter with the smooth density of $\mathcal{D}_b$, while $J_{\text{left},b}$ has the negative density and therefore the same zero locus. The occupied internal map is zero. Hence every full preconstraint point lies in the partial zero locus. A point of $\mathcal{C}_{\text{si}b}$ need not satisfy $\mathcal{H}_\perp = 0$, as already used to populate Proposition 11.2. $\square$

19. Elliptic orbit splitting and tame zero locus

Fix an auxiliary smooth metric and positive density only for the following Sobolev estimates and formal adjoints. The resulting orbit and slice subspaces are intrinsic.

Definition 19.1 (Orbit operator and slice direction). At $q \in \mathcal{Q}_{b,\text{fr}}^\infty$, define

$$R_q : \Gamma^\infty(T\Sigma) \longrightarrow T_q \mathcal{Q}_b^\infty, \quad R_q \xi = -\mathcal{L}_\xi q.$$

Let $R_q^\dagger$ be its formal L^2 adjoint and set

$$L_q = R_q^\dagger R_q, \quad S_q^\infty = \ker R_q^\dagger.$$

Proposition 19.2 (Elliptic tame splitting). *For the occupied vacuum and Klein–Gordon branches, L_q is a positive elliptic operator of order two whose kernel is the joint infinitesimal stabilizer. On the free locus it has a Green operator G_q , depending smoothly tamely on q , and*

$$T_q \mathcal{Q}_b^\infty = \text{im } R_q \oplus \mathcal{S}_q^\infty, \quad \mathbf{P}_{\text{orb},q} = R_q G_q R_q^\dagger.$$

For every $r \geq r_0$, the same formula extends boundedly as

$$R_q : H^{r+1}(T\Sigma) \rightarrow H^r(E_b), \quad G_q : H^{r-1}(T^*\Sigma) \rightarrow H^{r+1}(T\Sigma),$$

and supplies compatible closed Hilbert splittings.

Proof. For metrics, the principal symbol of R_q is the symmetrized covector product, hence $R_q^\dagger R_q$ is elliptic; its kernel is the Killing algebra. In the scalar branch the metric component is unchanged and the scalar Lie derivative adds a lower-order nonnegative term, so ellipticity and the kernel description persist. Trivial stabilizer therefore makes L_q invertible on compact Σ . Parametric elliptic regularity gives the Green family, the displayed Sobolev estimates, and smooth tame dependence on the smooth coefficients. The operator $\mathbf{P}_{\text{orb},q}$ is idempotent with image $\text{im } R_q$ and kernel $\ker R_q^\dagger$, proving both the tame inverse-limit splitting and its levelwise closures [2, 4]. $\square$

Proposition 19.3 (Split tame zero locus). *At every $x = (q, p) \in \mathcal{C}_{\text{si},b}^{\text{reg}}$, the differential of $J_{\text{si},b}$ is a tame split surjection onto the regular dual of the proper Lie algebra. Hence $\mathcal{C}_{\text{si},b}$ is a split tame Fréchet submanifold near x , and*

$$T_x \mathcal{C}_{\text{si},b} = \ker dJ_{\text{si},b}|_x.$$

Proof. For the occupied branches the joint map is the spatial map. In the vertical regular cotangent direction,

$$\langle dJ_{(q,p)}(0, \alpha), \xi \rangle = -\langle \alpha, \mathcal{L}_\xi q \rangle = \langle R_q^\dagger \alpha, \xi \rangle.$$

After the auxiliary L^2 identification, a tame right inverse is

$$\eta \longmapsto R_q G_q \eta, \quad R_q^\dagger R_q G_q \eta = \eta.$$

Thus dJ is tame split surjective throughout the declared free regular neighborhood. The tame submersion theorem gives the zero-locus statement. A conditional internal branch passes only when its $\mathfrak{C}_{\text{reg}}^2$ supplies the analogous joint elliptic complex, stabilizer, and tame right inverse; no such branch is reported occupied. $\square$

Lemma 19.4 (Moment-map orthogonality). *At every $x \in \mathcal{C}_{\text{si},b}^{\text{reg}}$,*

$$\ker dJ_{\text{si},b}|_x = (\text{im } \rho_{\text{prop},b}|_x)^{\Omega_b^\dagger}.$$

Proof. For every regular tangent v and proper parameter ϵ ,

$$\langle dJ_x(v), \epsilon \rangle = -\Omega_b^\dagger(\rho_x(\epsilon), v)$$

by Proposition 13.2 and its conditional internal analogue. This identity proves the displayed equality directly; no weak double-annihilator inference is used. $\square$

20. Slice-adapted characteristic calculation

Proposition 20.1 (Regular cotangent slice normal form). *For each $x \in \mathcal{C}_{\text{si},b}^{\text{reg}}$, there is an invariant neighborhood and a smooth tame configuration chart*

$$\mathcal{U}_q \cong \mathcal{V}_e \times S_{b,q}^\infty$$

with $\mathcal{V}_e \subset \text{Diff}_0^\infty(\Sigma)$. In the induced regular cotangent coordinates

$$(\varphi, s; \mu, \nu) \in T^\dagger(\mathcal{V}_e \times S_{b,q}^\infty),$$

the left-action moment map is the negative coadjoint transport of μ . Consequently $J = 0$ is $\mu = 0$, and

$$\Theta_b^\dagger|_{J=0} = \langle \nu, ds \rangle, \quad \ker \left(\Omega_b^\dagger|_{J=0} \right) = T_\varphi \mathcal{V}_e \oplus 0.$$

Proof. Proposition 19.2 and the smooth Ebin slice construction integrate the orbit/slice splitting to the stated local product. Left trivialization of its regular cotangent bundle writes the canonical one-form as

$$\Theta^\dagger = \langle \mu, \vartheta_L \rangle + \langle \nu, ds \rangle,$$

where ϑ_L is the left Maurer–Cartan form. The cotangent-lift moment map is the negative coadjoint transport of μ ; at zero it is exactly $\mu = 0$. Pullback to that locus removes both the μ -term and its differential, leaving the canonical regular slice-cotangent form. Its weak nondegeneracy makes the group coordinate the entire kernel. This is a coordinate calculation, not an appeal to a Fréchet double-annihilator theorem. $\square$

Proposition 20.2 (Characteristic equals orbit). *Let $\omega_{\mathcal{C}_{\text{sib}}} = \iota^* \Omega_b^\dagger$. At every $x \in \mathcal{C}_{\text{sib}}^{\text{reg}}$,*

$$\ker \omega_{\mathcal{C}_{\text{sib}}}|_x = \text{im } \rho_{\text{prop},b}|_x.$$

Proof. In the chart of Proposition 20.1, the tangent to the proper orbit is exactly $T_\varphi \mathcal{V}_e \oplus 0$, and the same subspace is the complete kernel of the restricted form. The equality is invariant under the chart transition maps. $\square$

21. Local descent map and completed reduced forms

Proposition 21.1 (Slice and quotient submersion). *For every $x \in \mathcal{C}_{\text{sib}}^{\text{reg}}$, there is an invariant neighborhood $\mathcal{U}_x$, a smooth tame slice $S_{b,x}^\infty$, and a smooth tame submersion*

$$\pi_{\text{prop},b} : \mathcal{U}_x \cap \mathcal{C}_{\text{sib}} \longrightarrow T^\dagger S_{b,x}^\infty$$

such that

$$\ker d\pi_{\text{prop},b}|_x = \text{im } \rho_{\text{prop},b}|_x.$$

*For every $r \geq r_0$, the smooth slice has a compatible Hilbert completion $S_{b,x}^r$, and $T^*S_{b,x}^r$ is the completed local quotient chart.*

Proof. In the coordinates of Proposition 20.1,

$$\pi_{\text{prop},b}(\varphi, s; 0, \nu) = (s, \nu).$$

It is a tame submersion and its vertical tangent is precisely the group direction. The levelwise elliptic splittings of Proposition 19.2 complete the slice to $S_{b,x}^r$. This construction never differentiates the full pullback action on one fixed $T^*Q_b^r$. $\square$

Proposition 21.2 (Reduced two-form). *There is a unique weakly symplectic smooth tame form $\omega_{\text{red},b}^\dagger$ on every regular local quotient satisfying*

$$\pi_{\text{prop},b}^* \omega_{\text{red},b}^\dagger = \iota^* \Omega_b^\dagger.$$

*Its extension to each $T^*S_{b,x}^r$ is the canonical strong Hilbert symplectic form, and the extensions commute with the Sobolev restriction maps.*

Proof. Proposition 20.1 identifies the reduced form with $d\langle\nu, ds\rangle$ on $T^\dagger S_{b,x}^\infty$, proving existence, uniqueness, and weak nondegeneracy. The cotangent bundle of the Hilbert slice $S_{b,x}^r$ carries its canonical strong form. All are defined by the same density pairing, so pullback along restriction agrees. $\square$

Theorem W2. 4 (Regular characteristic and local quotient theorem). For $b \in I_2^{\text{triv}} \amalg I_2^{\text{joint}}$ and every $x \in \mathcal{C}_{\text{sl}b}^{\text{reg}}$,

$$\boxed{\text{im } \rho_{\text{prop},b}|_x = \ker \omega_{\mathcal{C}_{\text{sl}b}}|_x = \ker d\pi_{\text{prop},b}|_x}.$$

A unique weakly symplectic smooth reduced form exists, with compatible strong Hilbert cotangent forms on all completed slice charts. The assertion is local to the populated free regular charts; rank-changing and nontrivial-isotropy points remain outside this equality unless a separate stratum certificate is supplied.

Proof. The first equality is Proposition 20.2; the second is Proposition 21.1; existence and uniqueness of the reduced form is Proposition 21.2. Propositions 19.2, 19.3, and 20.1 supply the tame split-rank and direct characteristic calculations. No fixed-index action or weak double-annihilator identity is used. $\square$

22. Singular strata and exact reportability

Proposition 22.1 (Groupoid retention of singular data). *Points with nontrivial stabilizer, rank change, or failure of a split slice remain objects of the global proper-redescription groupoid. They are not promoted to points of the regular local quotient presentations and are not discarded.*

Proof. The presentation cover includes declared neighborhoods of those phase points. Definition 25.4 then retains each point as an object and every admitted stabilizing canonical transformation as an isotropy arrow. Rank change is recorded by the failure of a regular slice coordinate, not by deletion from the object space. Calling the orbit set a manifold would conceal the failure; the action groupoid retains it without overclaiming a singular reduction theorem. $\square$

Part V

Groupoid-faithful global descent

23. Canonical continuation site

Definition 23.1 (Canonical continuation site). For a passing branch b , let

$$\mathcal{C}\text{Site}_b = (\text{Reg}_b, \text{Iface}_b, \text{Kappa}_b, \text{Res}_b, \text{Comp}_b, \text{Fix}_{\partial,b})$$

have:

1. objects $U \in \text{Reg}_b$ given by regular quotient neighborhoods, slice neighborhoods, and declared neighborhoods of singular objects;
2. admitted interfaces I_{UV} given by common invariant subneighborhoods or restriction-compatible continuations;
3. continuation morphisms κ_{UV} recording that the same WDW-1 branch and invariant profile can be compared on I_{UV} ;
4. ordinary restriction of canonical data and witnesses;
5. composition when the two interfaces admit a common restricted continuation; and
6. the compact-core boundary witness, or a populated branch-specific boundary certificate.

The continuation morphism types a comparison. It is not itself a canonical proper-redescription arrow.

Proposition 23.2 (Site closure). *C*Site_b is closed under identity restrictions, narrower admitted interfaces, and composition of compatible continuation chains.

Proof. Every local presentation compares to itself. Restricting a tower-compatible chart, slice, canonical field, constraint, source witness, or boundary witness to a smaller common domain preserves its type. If U compares to V and V to W through the same WDW-1 branch and a nonempty common restricted interface, the retained ancestry identifies a composite comparison from U to W . If no such interface exists, composition is undefined rather than forced. $\square$

24. Canonical preservation predicate

Definition 24.1 (Standing-bearing invariant family). For a local presentation ρ_U , define

$$\text{Inv}_b(\rho_U) = (\Omega_b^\dagger, J_{\text{si},b}, \mathcal{C}_{\text{si}b}, \text{Recon}_b, \text{ConnCert}_b, \text{BdyCert}_b, Q_{\partial,b}, \text{Sector}_b, \text{Src}\mathfrak{M}, \widehat{x}, \text{Att}_{\widehat{x}}, W_{\widehat{x}}, \lambda, \mathcal{O}_{\text{prop},b}),$$

with the source-model entries replaced by the certified zero-source class on the vacuum branch.

Definition 24.2 (Canonical preservation predicate). A candidate local arrow g_{UV} satisfies $\text{Pres}_b(g_{UV}; I_{UV})$ when:

1. it is the restriction of a proper spatial/internal canonical transformation;
2. it preserves $\Omega_b^\dagger$, $J_{\text{si},b} = 0$, and the regularity/orbit-type status on the interface;
3. reconstructed covariant data agree modulo the WDW-1 foliation-preserving presentation equivalence;
4. source class, attachment, witness, and pointed-lift identity agree strictly in their structured quotient;
5. boundary charge, flux, mapping-class, global-form, and registered sector data agree; and
6. every admitted observable has the same value.

Proposition 24.3 (Preservation is not circular). *The predicate of Definition 24.2 tests an already constructed canonical transformation against independently retained WDW-1 and boundary data; it does not declare a transformation proper because the transformation labels itself proper.*

Proof. Hamiltonianity and closure are established in Theorems W2.2–W2.3 before the predicate is formed. The WDW-1 source, reconstruction, connection, and boundary coordinates are imported independently. Properness is the vanishing profile of Definition 17.1. Thus every clause has an external witness and the conjunction is a verification step. $\square$

25. Canonical presentation and phase/action groupoids

Definition 25.1 (Presentation groupoid). $\mathbb{G}_{\text{pres}b} \rightrightarrows \text{Obj}_b$ has objects (U, ρ_U) , where $U \in \text{Reg}_b$ and ρ_U is a local canonical presentation. An arrow is a tuple

$$g_{UV} = (U, V, \kappa_{UV}, I_{UV}, \rho_U, \rho_V, \omega_{UV})$$

whose final coordinate witnesses $\text{Pres}_b(g_{UV}; I_{UV})$. Composition is defined only on compatible restricted interfaces.

Proposition 25.2 (Presentation-groupoid laws). $\mathbb{G}_{\text{pres}b}$ has identity, inverse, partial composition, associative composition where defined, neutral identities, and restriction compatibility.

Proof. The identity canonical transformation preserves every invariant. A proper canonical transformation has an inverse in the same connected proper action; vanishing charge and sector preservation are inverse-stable. Composition of compatible proper transformations is proper by Theorem W2.3 and preserves the invariant family transitively. Associativity is associativity of canonical maps, while restriction commutes with composition because all fields, transformations, and witnesses are restricted to the same interface. If boundary or sector data obstruct one operation, the corresponding arrow is absent rather than a failed groupoid law. $\square$

Definition 25.3 (Phase action, cover augmentation, and action anchor). Choose the presentation cover $\{U_a\}_{a \in A_b}$ used above and set

$$\widetilde{\mathcal{C}}_{\text{sib}} = \coprod_{a \in A_b} (U_a \cap \mathcal{C}_{\text{sib}}), \quad \mathbf{a}_b(a, z) = z, \quad \tau_b(a, z) = a.$$

$\mathbf{a}_b : \widetilde{\mathcal{C}}_{\text{sib}} \rightarrow \mathcal{C}_{\text{sib}}$ is the cover augmentation, while

$$\tau_b : \widetilde{\mathcal{C}}_{\text{sib}} \longrightarrow \text{Obj}(\mathbb{G}_{\text{presb}})$$

is the action anchor. If $g : a \rightarrow c$ is an arrow of $\mathbb{G}_{\text{presb}}$, define

$$\text{Act}_b(g, z) = g \cdot z$$

on the fibre product of presentation arrows with phase points in their admitted interface. The preservation predicate guarantees $\text{Act}_b(g, z) \in U_c \cap \mathcal{C}_{\text{sib}}$.

Definition 25.4 (Proper phase/action groupoid). Define

$$\mathbb{A}_{\text{propb}} = \mathbb{G}_{\text{presb}} \ltimes \widetilde{\mathcal{C}}_{\text{sib}} \rightrightarrows \widetilde{\mathcal{C}}_{\text{sib}}.$$

Its arrows are pairs (g, z) for which $\text{Act}_b(g, z)$ is defined, with

$$\begin{aligned} s(g, z) &= (a, z), & t(g, z) &= (c, g \cdot z), \\ (h, g \cdot z) \circ (g, z) &= (h \circ g, z), & (g, z)^{-1} &= (g^{-1}, g \cdot z). \end{aligned}$$

The identity at (a, z) is $(1_a, z)$.

Proposition 25.5 (Action-groupoid typing and orbit theorem). *The maps in Definition 25.4 are well-defined smooth tame structure maps on every regular chart and well-defined restriction maps on the global object. They satisfy compatibility with the action anchor τ_b , source/target compatibility, identity, inverse, associativity, and restriction. The $\mathbb{A}_{\text{propb}}$ -orbit of (a, z) , after applying $\mathbf{a}_b$, is exactly the proper spatial/internal orbit of z ; on the regular locus its tangent is $\text{im } \rho_{\text{prop}, b}|_z$.*

Proof. Definition 24.2 makes every presentation arrow an actual canonical map on its interface and makes $J = 0$ invariant. The source and target therefore land in the displayed object space. The equations follow from the presentation-groupoid laws and ordinary action associativity: $(h \circ g) \cdot z = h \cdot (g \cdot z)$. Restriction acts simultaneously on g and z . The defining fibre product is typed by $s_{\mathbb{G}_{\text{pres}}}(g) = \tau_b(a, z)$, and $\tau_b(c, g \cdot z) = t_{\mathbb{G}_{\text{pres}}}(g)$. Conversely, every proper transformation in a declared presentation interface determines an arrow pair (g, z) , so action-groupoid orbits and proper phase orbits coincide. Taking the derivative of the smooth tame action on the regular locus gives the stated tangent. $\square$

Definition 25.6 (Global quotient and universal property). The notation

$$\mathcal{C}_{\text{sib}} // \mathbb{A}_{\text{propb}}$$

denotes the Morita class of the action groupoid of Definition 25.4. A map from this quotient to an object Y is, by definition, a compatible family $f_a : U_a \cap \mathcal{C}_{\text{sib}} \rightarrow Y$ satisfying

$$f_c(g \cdot z) = f_a(z)$$

for every action arrow (g, z) . This invariant-map property, rather than an untyped orbit set, is the quotient universal property used below.

Proposition 25.7 (Regular local-completion compatibility). *On every free regular chart, the action-groupoid quotient is presented by $T^+S_{b,x}^\infty$, and its tangent-vertical distribution is $\ker d\pi_{\text{prop},b} = \text{im } \rho_{\text{prop},b}$. The completed spaces $T^*S_{b,x}^r$ form a compatible family of strong local slice completions. No completed Hilbert overlap map, and hence no global strong Hilbert atlas, is asserted. Nonfree and rank-changing points remain objects with their actual isotropy arrows.*

Proof. Proposition 21.1 identifies each regular action orbit with one fibre of the local projection. Proposition 25.5 identifies those fibres with the action-groupoid orbits, while Definition 25.6 gives smooth overlap compatibility before completion. The Sobolev completions and strong forms are those of Proposition 21.2; their completed overlap maps are not used. No regular local manifold presentation is assigned to a nonregular object. $\square$

26. Triple-interface coherence and path drift

Proposition 26.1 (Canonical triple coherence). *Let g_{UV}, g_{VW}, g_{UW} be proper arrows on a compatible triple interface. Then the direct and composite routes have the same invariant family:*

$$\text{Inv}_b(g_{UW} \cdot \rho_U) = \text{Inv}_b((g_{VW} \circ g_{UV}) \cdot \rho_U).$$

For every admitted phase point z , the composite route has the typed action arrow

$$(g_{VW}, g_{UV} \cdot z) \circ (g_{UV}, z) = (g_{VW} \circ g_{UV}, z).$$

If the direct arrow g_{UW} is strictly the same canonical map, the two targets agree. Otherwise the imported coherence witness is realized by an admitted action arrow between the two target presentations; their complete invariant families agree.

Proof. Both routes have the same source, target, and restricted branch. Canonical composition is associative, and every arrow preserves the symplectic, constraint, reconstruction, source, boundary, sector, and observable coordinates. Any residual difference invisible to this family is a local presentation choice. A visible difference would violate at least one arrow's preservation witness. Under the hypotheses, the direct and composite routes therefore have a coherence witness. $\square$

Proposition 26.2 (Holonomy/role-drift distinction). *Registered physical holonomy, curvature, mapping-class action, or topological-sector data are not prohibited by Proposition 26.1. Only unregistered path dependence of the same reconstruction, source, boundary, or standing coordinate is excluded.*

Proof. Registered holonomy and sector data occur explicitly in Inv_b . A path that changes them is not an arrow between the same proper-redescription objects; it is a standing-bearing transformation or a different sector. The coherence theorem compares only routes asserted to be proper arrows on the same fixed interface and invariant profile. It therefore does not erase physical holonomy. $\square$

27. Bridge to the Gauge Transition Groupoid Theorem

Definition 27.1 (Phase/action-to-role realization functor). Define

$$\mathcal{F}_b : \mathbb{A}_{\text{prop},b} \longrightarrow \mathbb{G}_{\text{role}}$$

by sending (a, z) to the boundary-fixed role-data presentation realized by the canonical phase point z in chart a , and (g, z) to the same interface, continuation morphism, canonical action, and preservation witness evaluated on the role invariant family. It sends restrictions, identities, inverses, and compatible composites to their role-level counterparts.

Proposition 27.2 (Clause-by-clause bridge). *The functor $\mathcal{F}_b$ instantiates the imported continuation-site, preservation, identity, inverse, partial-composition, associativity, restriction, triple-coherence, and obstruction-vanishing clauses.*

Proof. Definition 23.1 supplies the continuation site. Definition 24.2 supplies a typed preservation predicate with boundary fixation. Proposition 25.2 supplies identity, inverse, composition, associativity, neutrality, and restriction; Proposition 25.5 supplies the cover augmentation, action anchor, and action compatibility. Proposition 26.1 supplies the coherence witness. Since direct and composite canonical maps recover the same invariant family, the imported transition obstruction vanishes on every admitted triple. These assignments preserve source and target and commute with composition, so they define the stated functor. $\square$

Proposition 27.3 (No hidden selector). *The construction of $\mathbb{A}_{\text{propb}}$ and its regular local quotient presentations introduces no global representative selector.*

Proof. Each Ebin slice is defined only on a local invariant neighborhood. On overlaps, two slice representatives are related by arrows of $\mathbb{A}_{\text{propb}}$, and no rule chooses one slice across all objects. Singular objects and isotropy remain in the groupoid rather than being forced into a selected chart. A global section would be additional data and is nowhere used in the proof. $\square$

Theorem W2. 5 (Groupoid-faithful global descent theorem). For every WDW-2 passing branch, the canonical proper arrows instantiate the boundary-fixed local/restriction groupoid layer through the realization functor $\mathcal{F}_b$. The global descent object is

$$\mathcal{C}_{\text{sib}} // \mathbb{A}_{\text{propb}},$$

defined by the invariant-map universal property and equipped with its regular local symplectic quotient presentations and compatible strong local slice completions. It retains isotropy, singular strata, boundary and sector data, and admits no hidden global selector.

Proof. Propositions 23.2–25.7 construct the presentation and phase/action groupoids, their cover augmentation, action anchor, action, orbits, and universal quotient. Propositions 27.2 and 27.3 verify the imported bridge and rule out representative selection. Proposition 22.1 retains nonregular objects, and Proposition 17.3 retains standing-bearing transformations. The imported Boundary-Fixed Gauge Transition Groupoid-Layer Theorem and its Exhaustion of Weaker Realizations theorem then apply exactly at the role layer [7]; they do not supply the canonical moment map, which has already been proved independently. $\square$

Part VI

Source, observables, boundary, and the WDW-3 handoff

28. Complete-history groupoid

Definition 28.1 (History objects and arrows). For a canonical matter model $\mathfrak{M}$, let $\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M})$ be the WDW-1 complete smooth solution histories. $\mathbb{G}_{\text{histb}}$ has these histories as objects. An arrow is a smooth foliation-preserving path of proper spatial/internal transformations acting on every canonical variable and on lapse/shift by the transformation law required to reconstruct the same covariant fields. The arrow must preserve endpoint, boundary, and pointed-lift data.

Proposition 28.2 (Evaluation equivariance). *For every regular time t , the evaluation map*

$$\text{ev}_t : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) \longrightarrow \mathcal{C}_{\text{sib}}$$

is equivariant wherever the evaluated history lies in the declared spatial/internal zero locus.

Proof. The history arrow acts timewise by the same canonical transformation defining $\mathbb{A}_{\text{prop}b}$. Proper transformations preserve $J_{\text{si}} = 0$, so evaluating before or after the action gives points related by the corresponding phase-space arrow. The time-dependent contribution is absorbed in the standard shift transformation required by foliation-preserving reconstruction; it does not alter the canonical data at the evaluated leaf. $\square$

29. Source-realization-family descent

Proposition 29.1 (Historywise source invariance). *For every proper history arrow g and complete matter history Γ ,*

$$\text{Src}_{\mathfrak{M}}(g \cdot \Gamma) = \text{Src}_{\mathfrak{M}}(\Gamma).$$

The equality is in the structured Module-A source quotient and preserves attachment, witness, tensor representative up to its declared presentation equivalence, and pointed-lift identity.

Proof. WDW-1 constructs $\text{Src}_{\mathfrak{M}}$ by reconstructing the covariant matter fields and their metric response from a complete canonical history. A history arrow transforms all canonical variables, multipliers, and embedding presentation coherently; WDW-1 reconstruction therefore gives a foliation-preserving redescription of the same covariant solution. The Module-A source quotient is natural under that redescription. Definition 24.2 additionally requires the structured class, attachment, witness, and lift index to agree. Hence the source map is constant on every history orbit. $\square$

Corollary 29.2 (Descended source map). *There is a unique map*

$$\overline{\text{Src}}_{\mathfrak{M}} : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) // \mathbb{G}_{\text{hist}b} \longrightarrow \hat{\mathcal{Q}}_{\text{sm},0}^{\text{src}}$$

such that $\overline{\text{Src}}_{\mathfrak{M}} \circ \pi_{\text{hist}} = \text{Src}_{\mathfrak{M}}$.

Proof. Proposition 29.1 makes $\text{Src}_{\mathfrak{M}}$ constant on arrows, which is exactly the universal descent condition for the groupoid quotient. $\square$

Proposition 29.3 (Exact Klein–Gordon fibre). *For the WDW-1 distinguished history γ_{ϕ} ,*

$$\overline{\text{Src}}_{\phi}([\gamma_{\phi}]) = \hat{x}_{\phi}.$$

Every lawful pointed lift over $\hat{x}_{\phi}$ remains separately indexed.

Proof. WDW-1 proves $\text{Src}_{\phi}(\gamma_{\phi}) = \hat{x}_{\phi}$. By the defining factorization in Corollary 29.2, evaluation on the orbit class gives the same value. The quotient acts on presentation histories, not on the coproduct index of pointed lifts, so no two lawful lifts are silently identified. $\square$

30. Reduced observables

Definition 30.1 (Proper invariant observables). *On a regular neighborhood $\mathcal{U} \subset \mathcal{C}_{\text{si}b}^{\text{reg}}$, let*

$$\mathcal{O}_{\text{prop},b}(\mathcal{U}) = \{F \in C^{\infty}(\mathcal{U}) : \rho(\epsilon)F = 0 \text{ for every proper } \epsilon\}.$$

Proposition 30.2 (Local observable descent). *Pullback along $\pi_{\text{prop},b}$ is an isomorphism*

$$C^{\infty}(T^{\dagger}S_{b,x}^{\infty}) \longrightarrow \mathcal{O}_{\text{prop},b}(\mathcal{U}).$$

Smooth functions separate points of the local smooth tame slice chart and extend on each completed Hilbert slice chart.

Proof. A smooth quotient function pulls back to an invariant function. Conversely, an invariant smooth function is constant on the fibres of the quotient submersion, and the slice chart defines its unique smooth factor. Smooth local slice coordinates and their Hilbert completions provide separating functions. This proof is local and does not produce a global countable separating algebra on the possibly non-Hausdorff action-groupoid quotient. $\square$

31. Boundary and asymptotic extensions

Branch	Required WDW-2 data	Publication status
Compact core	$\partial\Sigma = \emptyset$; no surface term or flux	OCCUPIED
Finite boundary	boundary conditions, improved generator, charge and flux algebra	CONDITIONAL
Asymptotically flat	falloff, parity, ADM/BMS-sensitive charge profile	CONDITIONAL
AdS-like	boundary conditions and counterterm or equivalent charge prescription	CONDITIONAL
Null boundary	null action, corner data, and null charge formalism	TARGET UPDATE

Proposition 31.1 (No extension by analogy). *The occupied compact-core vanishing-charge result does not populate a finite, asymptotic, AdS-like, or null boundary branch.*

Proof. Each extension changes the differentiability domain, surface functional, flux law, and possibly the asymptotic symmetry algebra. These coordinates determine which transformations are proper. Their absence cannot be repaired by the empty-boundary calculation. $\square$

32. Large transformations and sector standing

Proposition 32.1 (Large-action retention). *The quotient by $\mathbb{A}_{\text{propb}}$ retains the action of the spatial mapping-class group and, on a certified internal branch, every disconnected internal component and registered topological-sector action.*

Proof. $\mathbb{A}_{\text{propb}}$ contains only arrows induced by the identity-component proper profile. Disconnected components do not become connected through local restriction. Their action is therefore an external symmetry or sector correspondence of the quotient groupoid, not an arrow collapsed inside it, unless a separate certificate proves that action to be pure presentation skin. $\square$

33. Normal-family equivariance

Proposition 33.1 (Spatial equivariance of the normal family). *On the smooth occupied domain,*

$$\{\mathcal{D}_b[\xi], H_{\perp,b}[N]\} = H_{\perp,b}[\mathcal{L}_\xi N].$$

Equivalently, for the declared left-action fundamental vector $\rho_{\text{sp},b}(\xi) = -X_{\mathcal{D}_b[\xi]}$,

$$\rho_{\text{sp},b}(\xi)(H_{\perp,b}[N]) = H_{\perp,b}[\mathcal{L}_\xi N].$$

On a populated internal branch,

$$\{\mathcal{G}_b[\alpha], H_{\perp,b}[N]\} = 0$$

with any declared boundary improvement included.

Proof. $\mathcal{H}_{\perp,b}$ is a scalar density of weight one constructed covariantly from the spatial canonical variables. The generator $\mathcal{D}[\xi]$ Lie-drags every field and density; integrating the Lie derivative of $N\mathcal{H}_{\perp}$ over compact Σ leaves the displayed transformation of the scalar smearing. Since $X_{\mathcal{D}[\xi]} = -\rho_{\text{sp}}(\xi)$, the second displayed equation is the same statement in the declared left-action convention. In the scalar model the internal factor is trivial. For a certified internal model, gauge invariance of the normal density is one of the Hamiltonian/action match coordinates and gives the second bracket. The result concerns the family $N \mapsto H_{\perp}[N]$: a fixed arbitrary $H_{\perp}[N]$ is not a basic function because N transforms. $\square$

Nonclaim 33.2 (Normal self-bracket). No use is made of a formula for $\{H_\perp[N], H_\perp[M]\}$. Its exact field-theoretic closure, characteristic role, boundary completion, and refoliation interpretation are the WDW-3 burden.

34. Exact WDW-3 handoff

Definition 34.1 (WDW-3 handoff). For every passing branch b , define

$$\begin{aligned} \mathfrak{H}_{W2}(b) = & (\mathcal{C}_{\text{sib}} // \mathbb{A}_{\text{prop}b}, \mathfrak{A}_b^{\text{reg}}, \overline{\mathcal{O}}_{\text{prop},b}, \overline{\text{Src}}_{\mathfrak{M}}, \overline{\text{Recon}}_b, \mathcal{H}_{\perp b}[\mathcal{N}], \\ & \text{Emb}_b, \text{BdyCert}_b, \text{Open}_{G_3,b}), \\ \mathfrak{A}_b^{\text{reg}} = & \{T^\dagger S_a^\infty, \omega_{\text{red},a}^\dagger, T^* S_a^r, \omega_{\text{red},a}^r\}_{a,r \geq r_0}. \end{aligned}$$

$\text{Open}_{G_3,b}$ records the Hamiltonian self-bracket, full normal characteristic kernel, refoliation, embedding comparison, and boundary normal-deformation burdens.

Proposition 34.2 (No slice substitution). *WDW-3 receives the groupoid object and the full family of regular local quotient presentations and slice completions, not one selected local slice or a claimed global Hilbert atlas. It receives the full equivariant lapse-smearing family, not a fixed-lapse function.*

Proof. One slice omits transition, isotropy, and singular data and would reintroduce the selector excluded by Proposition 27.3. A fixed lapse is not invariant under the spatial action by Proposition 33.1. The displayed handoff retains exactly the objects on which WDW-3 must establish the normal-deformation theorem. $\square$

Theorem W2. 6 (Source-history and observable descent theorem). Proper history arrows preserve the structured source class, attachment, witness, reconstruction class, and pointed-lift identity. The WDW-1 source map descends to the complete-history groupoid quotient, with

$$\overline{\text{Src}}_\phi([\gamma_\phi]) = \hat{x}_\phi.$$

On every regular local phase quotient, smooth proper-invariant observables are exactly pullbacks of smooth reduced observables.

Proof. Propositions 28.2–29.3 give history and source descent. Proposition 30.2 gives the local observable equivalence. The constructions keep the history quotient separate from the phase quotient. $\square$

Theorem W2. 7 (Normal-constraint equivariance and WDW-3 handoff theorem). The normal Hamiltonian-constraint family is equivariant under the completed proper spatial/internal groupoid, with its lapse smearing transformed. Definition 34.1 constructs the exact downstream object and records every unproved normal-deformation coordinate.

Proof. Proposition 33.1 proves equivariance. Nonclaim 33.2 prevents backfilling the WDW-3 result. Proposition 34.2 proves that the groupoid, source, reconstruction, boundary, and smearing coordinates survive the handoff. $\square$

35. WDW-2 capstone and gate ledger

Theorem W2. 8 (Source-realization-family-preserving groupoid-faithful proper-redescription descent with boundary-standing non-collapse). Let $b \in I_2^{\text{triv}} \amalg I_2^{\text{joint}}$ carry the field, spatial, internal when nontrivial, lift, properness, regularity, bridge, source-history when nonvacuum, and boundary certificates required by the frozen target. Then:

1. the spatial and admitted internal transformations possess a typed equivariant moment map;
2. the proper generator algebra closes on the common domain;

3. the spatial/internal zero locus is a defined split submanifold on every declared regular chart;
4. proper orbit, characteristic, and quotient-vertical directions coincide there;
5. a unique weakly symplectic local reduced form exists and has compatible strong Hilbert slice completions;
6. the global descent is the boundary-fixed phase/action groupoid with its invariant-map universal property;
7. no global representative selector is introduced;
8. every nonzero boundary, flux, large-transformation, and sector coordinate is retained;
9. the complete-history source-realization family descends unchanged;
10. local reduced observables and proper-invariant observables correspond;
11. the normal constraint is an equivariant lapse-smeared family; and
12. the branch-status and WDW-3 eligibility maps are single-valued.

The vacuum and attached Klein–Gordon branches occupy the spatial factor. The internal factor is asserted only for an independently populated I_2^{joint} branch.

Proof. Theorems W2.1–W2.3 construct the eligible carriers, Hamiltonian actions, proper algebra, and standing guard. Theorem W2.4 proves the local characteristic reduction. Theorem W2.5 constructs global groupoid descent. Theorem W2.6 supplies source and observable descent, and Theorem W2.7 supplies the exact normal-family handoff. Their hypotheses are precisely the certificate coordinates listed in the theorem. The occupied status follows from Proposition 6.2; the absence of a nontrivial internal occupant prevents a broader matter report. $\square$

Table 2: WDW-2 gate ledger.

Gate	Required result	Status	Exact report
G_1	W1.1–W1.8	IMPORTED VERIFIED	Final WDW-1 publication object; not reopened
G_{2a}	completed carrier/domain	PROVED	vacuum and KG
G_{2b}	spatial moment map	PROVED	vacuum and KG
G_{2c}	internal moment map	CONDITIONAL	no nontrivial occupant claimed
G_{2d}	proper closure	PROVED	compact spatial core
G_{2e}	characteristic equality	PROVED	free regular charts
G_{2f}	global groupoid descent	PROVED	all passing branches
G_{2g}	source descent	PROVED	KG complete histories
G_{2h}	boundary non-collapse	PROVED	compact core; extensions conditional
G_2	W2.8	PROVED	spatial occupied, internal certificate-gated
G_3	normal/refoliation closure	OUT OF SCOPE	exact handoff only

Part VII

Adversarial closure and publication standard

36. Adversarial verification map

The following table is part of the proof contract. An objection already answered here must identify a failed premise or inference in the cited response before it can reopen the gate.

Table 3: Adversarial verification map.

Attack	Exact response	Failure consequence
Undefined field phase space	Definitions 7.1–8.1 give the regular cotangent carrier, density pairing, Sobolev tower, and smooth domain.	G_{2a} fails if any map is absent.
Derivative loss is hidden	Proposition 8.2 records the one-level transport loss and prohibits a fixed-index action claim; Propositions 9.2 and 19.2 separate the smooth action from the Sobolev estimates.	W2.1–W2.4 fail.
Momentum constraint merely named	Lemmas 12.1–12.2 and Proposition 13.2 give the full variation.	W2.2 fails.
Moment-map sign conflicts with WDW-1	Definition 9.1, Proposition 13.2, and Remark 7.3 distinguish the conventional generator $\mathcal{D}$ from the declared left-action moment map $J_{\text{left}} = -\mathcal{D}$, proving $\iota_{\rho}\Omega = -dJ_{\text{left}} = d\mathcal{D}$.	Sign ledger fails if mixed.
Ordinary/covariant lifts conflated	Definition 14.1 and Proposition 14.2 record the Gauss correction and field dependence.	Joint branch blocked.
Nonsplit extension presumed split	Definition 16.1 uses the actual bundle-automorphism extension.	Internal theorem narrows.
Unresolved internal candidate called first class	Proposition 10.3 blocks it.	Branch is BLOCKED.
Central term ignored	The boundary/closure certificate requires its vanishing on the proper algebra.	Proper reduction fails or target changes.
Full Hamiltonian surface used too early	Definition 18.1 and Proposition 18.2 separate the partial zero locus.	Dependency violation.
Finite-dimensional reduction imported	Propositions 19.2–20.1 prove tame split rank and compute the characteristic kernel directly in slice coordinates.	W2.4 fails.
Stabilizers make quotient singular	The equality is on the free regular locus; Proposition 22.1 retains other objects.	Only a global-manifold claim is denied.
Naive orbit set called physical quotient	Definitions 25.1–25.6 construct the presentation groupoid, cover augmentation, action anchor, action map, action groupoid, and invariant-map universal property.	W2.5 fails.
Gauge slice is a hidden selector	Propositions 25.7 and 27.3 keep slices local and glue them by action arrows.	Descent inadmissible.
Gribov obstruction ignored	Multiple slices, overlap arrows, isotropy, and singular objects remain in the groupoid.	No global section is claimed.

Attack	Exact response	Failure consequence
Nonzero boundary charge quotiented	Proposition 17.3 excludes it from proper transformations.	Boundary theorem fails.
Large transformations collapsed	Proposition 32.1 retains mapping-class and sector actions.	Quotient overcollapses standing.
Quantum anomaly repairs classical closure	All brackets used here are classical and proved before any quantum gate.	Branch blocked.
Full phase space assigned to one source class	Definitions 28.1 and Corollary 29.2 define source only on complete histories.	W2.6 fails.
Proper arrow changes attachment/witness	Definition 24.2 and Proposition 29.1 preserve both.	Source faithfulness fails.
Fixed $H_{\perp}[N]$ called basic	Proposition 33.1 descends the family with transformed N .	W2.7 fails.
Hamiltonian self-bracket assumed	Nonclaim 33.2 makes it the exact WDW-3 burden.	Frozen-scope violation.
Global observable maximality claimed	Proposition 30.2 is explicitly local.	Claim narrows.
Groupoid theorem said to prove moment map	Theorem W2.2 precedes and is independent of the bridge in Theorem W2.5.	Invalid import.
Physical holonomy called role drift	Proposition 26.2 registers physical holonomy and excludes only unregistered drift.	Bridge fails if conflated.
Compact charge result generalized	Proposition 31.1 requires a certificate for every extension.	Extension removed.
Spatial reduction eliminates physical time	Theorem W2.7 passes normal and clock questions to WDW-3 and later work.	Time claim removed.

37. Failure conditions and falsifiers

Proposition 37.1 (Branchwise failure law). *A missing field, action, moment-map, closure, properness, regularity, bridge, source, or boundary certificate has exactly one of the following consequences: the branch is blocked; a local chart is removed; the internal theorem is removed; a boundary extension is removed; only the global groupoid is retained; or G_2 remains open. It does not license repair by a later quantum result.*

Proof. Each certificate is an antecedent of a named theorem. Removing it removes that theorem's conclusion on the affected branch. The branch partition and gate ledger record the maximal remaining result. None of the consequences changes the upstream WDW-1 object or reverses the dependency order. $\square$

38. Claim-status and terminology ledgers

Term	Fixed use
quotient	a regular local manifold quotient or the explicitly written groupoid quotient
proper	vanishing charge/flux and sector-preserving profile, not zero charge at one point
constraint	a defined component of a typed map, not a named geometric relation alone
moment map	a map satisfying the signed contraction identity on the declared carrier
equivariant family	$N \mapsto H_{\perp}[N]$ with the smearing transformed
basic function	invariant and horizontal fixed function; not asserted for arbitrary fixed N
source fibre	inverse image of a structured class under the complete-history source map
phase space	model-level regular cotangent carrier; never one fixed source fibre
global descent	the proper phase/action groupoid, not a presentation groupoid or presumed smooth orbit manifold

39. Publication synchronization

Audit condition 39.1 (Publication synchronization). The abstract, endpoint box, W2.8, conclusion, gate ledger, and adversarial map report the same branchwise endpoint. The source build reproduces that endpoint without changing its branch or scope statements.

Part VIII

Audit appendices

A. Convention and density audit

This appendix makes explicit every convention used by the WDW-2 calculation. It is part of the theorem object because a sign or density mismatch would change the generator, constraint zero locus, or source match.

A.1 Geometric conventions

$$\dim \Sigma = 3, \quad \text{sign}(g) = (-, +, +, +), \quad n^{\mu}n_{\mu} = -1,$$

$$h_{\mu\nu} = g_{\mu\nu} + n_{\mu}n_{\nu}, \quad K_{ij} = \frac{1}{2N}(\dot{h}_{ij} - \mathcal{L}_{\vec{N}}h_{ij}).$$

The WDW-1 gravitational momentum is

$$\pi^{ij} = \frac{\sqrt{h}}{2\kappa}(K^{ij} - Kh^{ij}),$$

a contravariant symmetric tensor density of weight one. The scalar momentum is

$$p_{\phi} = \frac{\sqrt{h}}{N}(\dot{\phi} - N^i D_i \phi) = \sqrt{h} n^{\mu} \nabla_{\mu} \phi,$$

a scalar density of weight one.

A.2 Canonical pairing and sign

For a configuration q and density momentum p ,

$$\Theta(\delta q, \delta p) = \langle p, \delta q \rangle, \quad \Omega = d\Theta.$$

Thus

$$\Omega((u, \alpha), (v, \beta)) = \langle \alpha, v \rangle - \langle \beta, u \rangle.$$

WDW-1 fixes

$$\iota_{X_F} \Omega = -dF.$$

In finite coordinates this is

$$X_F = \frac{\delta F}{\delta p} \frac{\delta}{\delta q} - \frac{\delta F}{\delta q} \frac{\delta}{\delta p},$$

and the Poisson bracket is

$$\{F, G\} = dF(X_G) = \left\langle \frac{\delta F}{\delta q}, \frac{\delta G}{\delta p} \right\rangle - \left\langle \frac{\delta F}{\delta p}, \frac{\delta G}{\delta q} \right\rangle.$$

For $\mathcal{D}[\xi] = \langle p, \mathcal{L}_\xi q \rangle$, the Hamiltonian flow is ordinary positive Lie dragging, while the declared left group action $\varphi \cdot q = (\varphi^{-1})^* q$ has infinitesimal $-\mathcal{L}_\xi q$. The signed left-action moment map and its signed equation are consequently

$$\langle J_{\text{left}}, \xi \rangle = -\mathcal{D}[\xi], \quad \iota_{\rho(\xi)} \Omega = d\mathcal{D}[\xi] = -d\langle J_{\text{left}}, \xi \rangle.$$

Thus $X_{\mathcal{D}[\xi]} = -\rho(\xi)$. The zero loci of $\mathcal{D}$ and J_{left} coincide, but their signs are not interchangeable in the moment-map identity.

A.3 Density table

Object	Weight/type	Pairing or transformation
h_{ij}	covariant symmetric tensor, weight 0	$\mathcal{L}_\xi h_{ij} = D_i \xi_j + D_j \xi_i$
π^{ij}	contravariant symmetric density, weight 1	$\int \pi^{ij} \delta h_{ij}$
ϕ	scalar, weight 0	$\mathcal{L}_\xi \phi = \xi^i D_i \phi$
p_ϕ	scalar density, weight 1	$\int p_\phi \delta \phi$
$\mathcal{H}_i$	covector density, weight 1	$\int \xi^i \mathcal{H}_i$
$\mathcal{H}_\perp$	scalar density, weight 1	$\int N \mathcal{H}_\perp$
N	scalar smearing	$\delta_\rho N = -\mathcal{L}_\xi N; \{\mathcal{D}[\xi], H_\perp[N]\} = H_\perp[\mathcal{L}_\xi N]$
ξ^i	vector parameter	Lie bracket $[\xi, \eta]$

A.4 Source sign match

WDW-1 defines

$$\rho = T_{\mu\nu} n^\mu n^\nu, \quad j_i = -h_i^\mu n^\nu T_{\mu\nu}.$$

For the minimally coupled scalar,

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} (\nabla_\alpha \phi \nabla^\alpha \phi + m^2 \phi^2).$$

Therefore

$$\rho_\phi = \frac{1}{2} \left[\left(\frac{p_\phi}{\sqrt{h}} \right)^2 + D_i \phi D^i \phi + m^2 \phi^2 \right], \quad j_{i,\phi} = -\frac{p_\phi}{\sqrt{h}} D_i \phi,$$

and

$$\begin{aligned} \mathcal{H}_\perp^\phi &= \frac{p_\phi^2}{2\sqrt{h}} + \frac{\sqrt{h}}{2} (D_i \phi D^i \phi + m^2 \phi^2) = \sqrt{h} \rho_\phi, \\ \mathcal{H}_i^\phi &= p_\phi D_i \phi = -\sqrt{h} j_{i,\phi}. \end{aligned}$$

This synchronizes the spatial generator with the WDW-1 source projection.

B. Field phase-space carrier and scale audit

B.1 Regular cotangent geometry

The configuration spaces

$$\text{Met}^\infty(\Sigma), \quad \text{Met}^\infty(\Sigma) \times C^\infty(\Sigma)$$

are tame Fréchet manifolds. Their regular cotangent fibres are smooth density-valued dual sections:

$$\begin{aligned} \mathcal{P}_g^\dagger &= \text{Met}^\infty(\Sigma) \times \Gamma^\infty(S^2 T^* \Sigma \otimes \text{Dens}), \\ \mathcal{P}_\phi^\dagger &= \mathcal{P}_g^\dagger \times C^\infty(\Sigma) \times \Gamma^\infty(\text{Dens}). \end{aligned}$$

The word “regular” is essential. No claim is made that these fibres exhaust the continuous dual of a Fréchet tangent space. The intrinsic density pairing defines $\Theta^\dagger$ and a weakly symplectic $\Omega^\dagger = d\Theta^\dagger$.

B.2 Sobolev tower and the admitted loss

Choose an auxiliary smooth metric only to form norms. For $r \geq r_0 > 7/2$,

$$H^r(S^2 T^* \Sigma) \hookrightarrow C^2(S^2 T^* \Sigma),$$

so $\text{Met}^r(\Sigma)$ is a Hilbert manifold and smooth positive metrics are dense. The levelwise Lie derivative has the typed estimate

$$\|\mathcal{L}_\xi q\|_{H^r} \leq C_r(q) \|\xi\|_{H^{r+1}} \|q\|_{H^{r+1}}.$$

It is therefore a map from level $r+1$ to level r , not a tangent map on the entire level- r configuration manifold. The familiar counterexample

$$q \in H^r \setminus H^{r+1}, \quad \xi = \partial_1$$

on a torus gives $\mathcal{L}_\xi q \notin H^r$ in general. This is why no Diff^{r+1} -smooth action or full-dual moment map is asserted on one fixed $T^* \mathcal{Q}_b^r$.

The tower is instead used through

$$\mathcal{Q}_b^\infty = \bigcap_{r \geq r_0} \mathcal{Q}_b^r$$

with the projective-limit topology. Smooth pullback and Lie differentiation remain in every level simultaneously, while the loss estimates provide the tame bounds.

B.3 Smooth WDW-1 identification

For a smooth momentum density π ,

$$\ell_\pi(k) = \int_\Sigma \pi^{ij} k_{ij}$$

is the exact WDW-1 canonical pairing. The analogous scalar pairing is $\ell_p(\psi) = \int_{\Sigma} p_{\phi} \psi$. These formulas identify the WDW-1 smooth momenta with the regular cotangent fibres without adding distributional covectors. Thus

$$(\iota_b^{\dagger})^* \Theta_b^{\dagger} = \Theta_b^{W1}, \quad (\iota_b^{\dagger})^* \Omega_b^{\dagger} = \Omega_b^{W1}.$$

The configuration projections to every Sobolev level are dense comparison maps, not replacement phase carriers.

B.4 Smooth action and moment map

For $\varphi \in \text{Diff}_0^{\infty}(\Sigma)$,

$$\varphi \cdot (q, p) = ((\varphi^{-1})^* q, (\varphi^{-1})^* p).$$

Both entries are smooth tensor or tensor-density pullbacks. The action is smooth tame and preserves the density pairing. Its left-action moment map is

$$\langle J_{\text{left}}(q, p), \xi \rangle = -\langle p, \mathcal{L}_{\xi} q \rangle$$

with values in smooth one-form densities. Its differential is the fully typed expression

$$\langle dJ_{\text{left},(q,p)}(u, \alpha), \xi \rangle = -\langle \alpha, \mathcal{L}_{\xi} q \rangle - \langle p, \mathcal{L}_{\xi} u \rangle.$$

Every entry is smooth. The vertical component is the formal adjoint of the elliptic orbit operator and supplies the tame right inverse used in Proposition 19.3.

B.5 Reduced Hilbert completions

At a free smooth configuration, the elliptic projections

$$P_{\text{orb},q} = R_q G_q R_q^{\dagger}, \quad P_{\text{sl},q} = 1 - P_{\text{orb},q}$$

extend boundedly at every Sobolev level. The image of $P_{\text{sl},q}$ integrates to a compatible Hilbert slice $S_{b,q}^r$. Only after the orbit coordinate and its conjugate zero-momentum coordinate have been removed do we form

$$T^* S_{b,q}^r.$$

This Hilbert cotangent bundle has its canonical strong symplectic form. Hence strong symplecticity is a reduced-chart statement, not an invalid assertion about the unreduced fixed-index action.

B.6 Why the normal family remains smooth-domain data

The ADM normal density contains scalar curvature and nonlinear products of momenta. WDW-2 therefore retains

$$H_{\perp} : \mathcal{D}_b \times \mathcal{N} \rightarrow \mathbb{R}$$

on the exact WDW-1 smooth domain. Its spatial covariance is proved there. No normal Hamiltonian vector field on a Sobolev completion is required.

Certificate B.1 (Field carrier coordinates). For the occupied branches,

b	$\mathcal{Q}_b^{\infty}$	$\mathcal{P}_b^{\dagger}$	strong reduced completions
g	Met^{∞}	$T^{\dagger} \text{Met}^{\infty}$	$\{T^* S_{g,x}^r\}_{r \geq r_0}$
ϕ	$\text{Met}^{\infty} \times C^{\infty}$	$T^{\dagger}(\text{Met}^{\infty} \times C^{\infty})$	$\{T^* S_{\phi,x}^r\}_{r \geq r_0}$

The unreduced forms are weakly symplectic; the completed reduced slice forms are strong.

B.7 Branchwise ancestry audit

The vacuum branch retains

$$(\hat{x}_0, \mathcal{P}_g^\infty, \Omega_g, \Phi_g, \mathcal{C}_{\text{vac}}, \mathcal{N}, \mathcal{N}_+, \mathcal{X}, \text{Recon}_{\hat{x}_0}, \text{ConnCert}, \text{BdyCert}).$$

The Klein–Gordon branch retains every pointed lift

$$\lambda_\phi = (\mathfrak{M}_\phi, \text{Src}_\phi, \gamma_\phi, c_{\lambda_\phi}) \in \mathfrak{L}^+(\hat{x}_\phi).$$

The source map remains

$$\text{Src}_\phi : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}_\phi) \rightarrow \hat{Q}_{\text{sm},0}^{\text{src}}.$$

It is not extended to the model phase space or to any Sobolev completion. The carrier and tower constructions neither create source classes for phase points nor remove the structured ancestry of smooth histories.

B.8 Auxiliary-topology independence

If two auxiliary smooth metrics define the Sobolev tower, compactness makes their norms levelwise equivalent. The density pairing itself uses neither auxiliary volume form. Changing the modeling metric therefore changes estimates and chart presentation but not $\Theta^\dagger$, $\Omega^\dagger$, J , the zero locus, the action groupoid, or the source quotient.

C. Spatial generator variation

C.1 Gravitational term

Begin from the canonical generator before integration by parts:

$$\mathcal{D}_g[\xi] = \int_{\Sigma} \pi^{ij} \mathcal{L}_\xi h_{ij}.$$

Metric compatibility and symmetry give

$$\mathcal{L}_\xi h_{ij} = \xi^k D_k h_{ij} + h_{kj} D_i \xi^k + h_{ik} D_j \xi^k = D_i \xi_j + D_j \xi_i.$$

Hence

$$\begin{aligned} \mathcal{D}_g[\xi] &= 2 \int_{\Sigma} \pi^{ij} D_i \xi_j \\ &= -2 \int_{\Sigma} \xi_j D_i \pi^{ij} + 2 \int_{\partial\Sigma} r_i \pi^{ij} \xi_j \\ &= \int_{\Sigma} \xi^i (-2 D_j \pi^j_i) + B_g[\xi]. \end{aligned}$$

The core has $B_g = 0$.

Varying the pre-integration form is the safest route:

$$\delta \mathcal{D}_g[\xi] = \int_{\Sigma} \delta \pi^{ij} \mathcal{L}_\xi h_{ij} + \pi^{ij} \mathcal{L}_\xi \delta h_{ij}.$$

Using that the integral of a Lie derivative of a weight-one scalar density vanishes on compact Σ ,

$$\int_{\Sigma} \pi^{ij} \mathcal{L}_\xi \delta h_{ij} = - \int_{\Sigma} (\mathcal{L}_\xi \pi^{ij}) \delta h_{ij}.$$

Therefore

$$\delta \mathcal{D}_g[\xi] = \int_{\Sigma} \left(\delta \pi^{ij} \mathcal{L}_\xi h_{ij} - (\mathcal{L}_\xi \pi^{ij}) \delta h_{ij} \right).$$

The Hamiltonian vector field is

$$X_{\mathcal{D}_g[\xi]}(h, \pi) = (\mathcal{L}_\xi h, \mathcal{L}_\xi \pi),$$

exactly ordinary Lie dragging in the WDW-1 convention.

C.2 Scalar term

$$\mathcal{D}_m[\xi] = \int_{\Sigma} p_{\phi} \xi^i D_i \phi.$$

Its variation is

$$\delta \mathcal{D}_m[\xi] = \int_{\Sigma} (\delta p_{\phi} \mathcal{L}_{\xi} \phi + p_{\phi} \mathcal{L}_{\xi} \delta \phi).$$

Since $p_{\phi} \delta \phi$ is a weight-one density,

$$\int_{\Sigma} p_{\phi} \mathcal{L}_{\xi} \delta \phi = - \int_{\Sigma} (\mathcal{L}_{\xi} p_{\phi}) \delta \phi$$

on the compact core. Hence

$$X_{\mathcal{D}_m[\xi]}(\phi, p_{\phi}) = (\mathcal{L}_{\xi} \phi, \mathcal{L}_{\xi} p_{\phi}).$$

Adding gravity gives the cotangent lift on the full scalar model.

C.3 Bracket

Apply the first generator to the second:

$$\{\mathcal{D}[\xi], \mathcal{D}[\eta]\} = X_{\mathcal{D}[\eta]} \mathcal{D}[\xi].$$

Naturality of the pairing gives

$$X_{\mathcal{D}[\eta]} \langle p, \mathcal{L}_{\xi} q \rangle = \langle \mathcal{L}_{\eta} p, \mathcal{L}_{\xi} q \rangle + \langle p, \mathcal{L}_{\xi} \mathcal{L}_{\eta} q \rangle.$$

After moving $\mathcal{L}_{\eta}$ off the density pairing, the first term is $-\langle p, \mathcal{L}_{\eta} \mathcal{L}_{\xi} q \rangle$. Thus

$$\{\mathcal{D}[\xi], \mathcal{D}[\eta]\} = \langle p, [\mathcal{L}_{\xi}, \mathcal{L}_{\eta}] q \rangle = \mathcal{D}[[\xi, \eta]].$$

This derivation uses no Hamiltonian constraint.

C.4 Finite boundary

If a boundary is present, the above integrations produce

$$B_g[\xi] = 2 \int_{\partial \Sigma} r_i \pi^{ij} \xi_j,$$

plus model-specific matter or internal terms. The variation of the total generator is differentiable only on a chosen boundary phase space and after a charge improvement. The proper condition is not $Q_{\partial}[\xi](z_0) = 0$ at one point; it is

$$Q_{\partial}[\xi] \equiv 0, \quad \delta Q_{\partial}[\xi] \equiv 0, \quad \text{Flux}_{\partial}[\xi] = 0$$

on the whole admitted boundary profile.

D. Internal generator and resolved Dirac ledger

D.1 Conditional model-independent form

Let a certified internal branch have configuration variables Q^A , momenta P_A , and infinitesimal internal action $R_{\alpha} Q^A$. Before integration by parts its cotangent generator is

$$\mathcal{G}[\alpha] = \int_{\Sigma} P_A R_{\alpha} Q^A + Q_{\partial}^{\text{int}}[\alpha].$$

The variation is

$$\delta \mathcal{G}[\alpha] = \int_{\Sigma} (\delta P_A R_{\alpha} Q^A + P_A D R_{\alpha}(Q) \delta Q^A) + \delta Q_{\partial}^{\text{int}}[\alpha].$$

The certificate must show that integrations by parts isolate a bulk Gauss density, cancel the surface variation, and define a Hamiltonian vector field. If α is field-dependent, its variation is included before the Poisson bracket is taken.

D.2 Second-class resolution requirement

Suppose the Dirac ledger contains constraints χ_a with invertible matrix

$$C_{ab}(x, y) = \{\chi_a(x), \chi_b(y)\}.$$

Then the internal certificate must either solve $\chi_a = 0$ and pull back the symplectic form or use the corresponding Dirac bracket

$$\{F, G\}_D = \{F, G\} - \iint \{F, \chi_a(x)\} C^{ab}(x, y) \{\chi_b(y), G\} dx dy.$$

Only after this operation may remaining first-class Gauss directions be included in $\mathbb{A}_{\text{prop}}$. A label “Gauss candidate” without this resolution is BLOCKED.

D.3 Occupied scalar ledger

Coordinate	Klein–Gordon status	Reason
Matter Legendre map	REGULAR	$p_\phi = \sqrt{\hbar} n^\mu \nabla_\mu \phi$ uniquely determines the normal derivative
Matter primary constraints	NONE	Hessian in the scalar velocity is nonzero
Matter Gauss constraints	NONE	no internal gauge action
Matter second-class sector	NONE	no primary matter constraint pair
Matter reducibility	NONE	no internal generator
Internal carrier $\mathcal{K}_\phi$	0	exact WDW-1 scalar ledger
Spatial moment map	PROVED	$p_\phi D_i \phi$
Normal self-bracket	OUT OF SCOPE	WDW-3 burden

The vacuum ledger is the same with the scalar rows absent. No nontrivial internal ledger is reported as occupied.

E. Ordinary versus gauge-covariant diffeomorphisms

For an internal connection A , Cartan’s identity in covariant form reads schematically

$$\mathcal{L}_\xi A = \iota_\xi F + D(\iota_\xi A).$$

The ordinary Lie drag therefore differs from the gauge-covariant spatial action $\iota_\xi F$ by an internal transformation with parameter $\iota_\xi A$. At generator level,

$$\mathcal{D}^{\text{cov}}[\xi] = \mathcal{D}^{\text{ord}}[\xi] + \mathcal{G}[\iota_\xi A]$$

in the convention used in the main text.

The field dependence matters:

$$\delta(\iota_\xi A) = \iota_\xi \delta A.$$

Consequently

$$\{\mathcal{G}[\iota_\xi A], F\}$$

has the ordinary Gauss action plus the variation of its parameter whenever F acts on A . A joint algebra written with $\mathcal{D}^{\text{cov}}$ must retain that term. The orbit distribution after adjoining all Gauss directions is unchanged, but the algebraic presentation is not. This is why the conditional internal certificate stores

$$(\text{Lift}_\Sigma, \mathcal{D}^{\text{ord}}, \mathcal{D}^{\text{cov}}, \Delta_{\text{Gauss}}, \text{JointBracket})$$

as a single coordinate.

F. Boundary generator calculations

F.1 Compact core

$$\partial\Sigma = \emptyset \implies B_\partial = Q_\partial = \delta Q_\partial = \text{Flux}_\partial = 0.$$

This is the sole occupied boundary branch. It proves differentiability of the spatial generator without a surface improvement.

F.2 Finite boundary schema

For a finite spatial boundary $S = \partial\Sigma$, write $G_{\text{left}}^{\text{bulk}}[\epsilon]$ for the declared left-action moment-map functional. Its variation has the form

$$\delta G_{\text{left}}^{\text{bulk}}[\epsilon] = -\iota_{\rho(\epsilon)}\Omega + \int_S \mathcal{B}_\epsilon(\delta z).$$

The conventional positive-Lie-drag generator has the opposite bulk sign. A boundary certificate fixes one of these equivalent descriptions and carries that sign through its improvement and charge. A boundary certificate must specify:

1. allowed variations $\delta z|_S$;
2. allowed parameters $\epsilon|_S$;
3. a functional $Q_S[\epsilon]$ with $\delta Q_S = -\int_S \mathcal{B}_\epsilon$;
4. flux through S ;
5. the Poisson algebra and any cocycle; and
6. whether the transformation changes a boundary observable.

Only the profile with identically zero charge, variation, flux, and sector action is proper.

F.3 Asymptotically flat schema

An asymptotically flat branch additionally requires falloff and parity data. The improved Hamiltonian generally has ADM surface charges. A nonzero asymptotic translation or rotation therefore cannot be quotiented merely because the bulk constraint vanishes. The asymptotic symmetry algebra, possible supertranslations, and any central term are branch data. None are imported into the compact theorem.

F.4 AdS-like and null schemas

AdS-like targets require boundary conditions and counterterms or an equivalent covariant charge prescription. Null boundaries require a different variational principle, corner terms, and a null charge/flux law. These are target updates. The global groupoid theorem continues to provide the correct non-collapse grammar, but occupation requires the missing canonical calculations.

G. Constant-rank and slice proof

G.1 Infinitesimal orbit map

At $q \in \mathcal{Q}_{b,\text{fr}}^\infty$, define

$$R_q : \Gamma^\infty(T\Sigma) \longrightarrow T_q \mathcal{Q}_b^\infty, \quad R_q \xi = -\mathcal{L}_\xi q.$$

Trivial stabilizer makes R_q injective. Its principal symbol is injective because the metric component is the symmetrized covector product. Thus

$$L_q = R_q^\dagger R_q$$

is positive elliptic of order two; the scalar component adds only a lower-order nonnegative term. Its Green operator gives

$$T_q \mathcal{Q}_b^\infty = \text{im } R_q \oplus \ker R_q^\dagger.$$

At every $r \geq r_0$,

$$R_q : H^{r+1}(T\Sigma) \rightarrow H^r(E_b), \quad L_q : H^{r+1}(T\Sigma) \rightarrow H^{r-1}(T^*\Sigma)$$

and G_q has the reverse mapping order. The projections $R_q G_q R_q^\dagger$ and $1 - R_q G_q R_q^\dagger$ are bounded at every level and compatible under restriction.

G.2 Cotangent moment map differential

For the cotangent lift,

$$\langle J_{\text{left}}(q, p), \xi \rangle = -\langle p, \mathcal{L}_\xi q \rangle.$$

At $(q, 0)$, variation in the covector direction gives

$$\langle dJ_{\text{left},(q,0)}(0, \alpha), \xi \rangle = -\langle \alpha, \mathcal{L}_\xi q \rangle = \langle R_q^* \alpha, \xi \rangle.$$

Here R_q^* denotes the regular-dual realization of the formal adjoint $R_q^\dagger$. For a smooth one-form density η , set

$$\mathbf{K}_q \eta = R_q G_q \eta.$$

Then $R_q^\dagger \mathbf{K}_q \eta = \eta$. Parametric elliptic regularity makes $\mathbf{K}_q$ smooth tame in q . Thus dJ is a tame split surjection on the declared free regular neighborhood, and the tame submersion theorem gives the split zero locus.

G.3 Local product

The smooth slice theorem gives a local tame diffeomorphism

$$\mathcal{V}_e \times S_q^\infty \longrightarrow \mathcal{U}_q, \quad (\varphi, s) \longmapsto \varphi \cdot s$$

on the free locus. Regular cotangent lifting and zero-momentum reduction give the local model

$$J^{-1}(0) \cap T^\dagger \mathcal{U}_q \cong \mathcal{V}_e \times T^\dagger S_q^\infty.$$

The quotient is locally $T^\dagger S_q^\infty$, and the reduced form is its canonical weak form. Completing the slice at level r gives $T^* S_q^r$ with its canonical strong Hilbert form. This gives the concrete proof of Proposition 21.2 without a full finite-level group action.

G.4 Nonfree strata

If the stabilizer is a fixed compact subgroup H , the slice has an H -action and the local quotient is a stratified quotient of the slice. WDW-2 retains the corresponding isotropy arrows but does not report the free-manifold equality there without an independent stratum certificate. Rank-changing points remain only in the global groupoid.

H. Characteristic equality line by line

Fix a passing branch and a point $x \in \mathcal{C}_{\text{si}}^{\text{reg}}$. Abbreviate

$$V = T_x \mathcal{P}^\dagger, \quad W = \text{im } \rho_x, \quad K = \ker dJ_x.$$

The following steps prove the equality without a weak double-annihilator shortcut.

Step 1: $K = T_x \mathcal{C}_{\text{si}}$

The elliptic Green operator constructs a tame right inverse for dJ_x . The tame submersion theorem therefore makes $J^{-1}(0)$ a split tame Fréchet submanifold near x , and its tangent is the kernel of dJ_x .

Step 2: $K = W^\Omega$

For $v \in V$ and every proper parameter ϵ ,

$$\langle dJ_x(v), \epsilon \rangle = -\Omega_x(\rho_x(\epsilon), v).$$

Thus $dJ_x(v) = 0$ if and only if $\Omega(w, v) = 0$ for every $w \in W$.

Step 3: slice-adapted cotangent coordinates

In the local product $\mathcal{V}_e \times S_x^\infty$, left-trivialized regular cotangent coordinates are $(\varphi, s; \mu, \nu)$, with

$$\Theta^\dagger = \langle \mu, \vartheta_L \rangle + \langle \nu, ds \rangle.$$

The left-action moment map is the negative coadjoint transport of μ . Thus $J = 0$ is exactly $\mu = 0$.

Step 4: $\ker \omega_{\mathcal{C}_{\text{si}}} = W$

On $\mu = 0$,

$$\Theta^\dagger|_{\mathcal{C}_{\text{si}}} = \langle \nu, ds \rangle.$$

The slice cotangent form is weakly nondegenerate in (s, ν) , while the group coordinate φ does not appear. Therefore the complete kernel is the group tangent:

$$\ker \omega_{\mathcal{C}_{\text{si}}, x} = T_\varphi \mathcal{V}_e = W.$$

Step 5: $\ker d\pi = W$

The local quotient map is a submersion whose fibres are the proper group orbits. Its vertical tangent is therefore the orbit tangent:

$$\ker d\pi_x = W.$$

Step 6: descent and nondegeneracy

The quotient projection is

$$(\varphi, s; 0, \nu) \mapsto (s, \nu).$$

Thus the reduced form is the canonical weak form on $T^\dagger S_x^\infty$, and

$$\pi^* \omega_{\text{red}} = \omega_{\mathcal{C}_{\text{si}}}.$$

At every Sobolev level it extends to the canonical strong form on $T^* S_x^r$. The density formula makes those extensions compatible.

Collecting Steps 4 and 5 gives

$$\text{im } \rho_x = \ker \omega_{\mathcal{C}_{\text{si}}, x} = \ker d\pi_x.$$

I. Gauge Transition Groupoid import

The canonical bridge imports role-level structure from *The Gauge Transition Groupoid Theorem: Boundary-Fixed Local Presentation Rigidity under Realized Physical Admissibility* [7]. Its source role is recorded in Section 3.

Table 4: Exact groupoid import map.

Imported label	Hypotheses used	WDW-2 instantiation	Prohibited promotion
Definition “Continuation site” (<code>def:site</code>)	admitted regions, interfaces, continuation morphisms, restriction, composition, boundary fixation	Definition 23.1	does not supply Sobolev topology or dynamics
Definition “Preservation predicate” (<code>def:preserve</code>)	typed arrows and a fixed invariant family	Definitions 24.1–24.2	does not prove an arrow Hamiltonian
Identity, inverse, composition, and associativity propositions	admissible arrows and compatible interfaces	Proposition 25.2	does not prove the canonical bracket
Restriction propositions (<code>prop:restradm</code> ; <code>prop:</code> <code>restriction</code>)	boundary witnesses survive narrowing	ordinary restriction in the canonical site	does not assert a global section
Proposition “Triple-interface coherence” (<code>prop:triple</code>)	direct and composite routes are admitted on one fixed interface	Proposition 26.1	does not erase registered holonomy
Theorem “Gauge realization stability” (<code>thm:</code> <code>obstruction</code>)	same source, target, interface, and invariant family	vanishing canonical role drift	does not forbid physical curvature
Theorem “Boundary-fixed gauge transition groupoid-layer” (<code>thm:groupoid</code>)	continuation site and witnessed preservation predicate	realization functor $\mathcal{F}_b$	does not imply a global smooth orbit manifold
Theorem “Exhaustion of weaker gauge-transition realizations” (<code>thm:exhaustion</code>)	candidate local transition regime preserves the invariant family	no-selector and failure classification	does not classify dynamics or measurements

1.1 Why the bridge is substantive

The role theorem alone does not know:

$$\mathcal{P}_b^\dagger, \quad \Omega_b^\dagger, \quad \mathcal{D}_b, \quad J_{\text{si},b}, \quad \mathcal{C}_{\text{si},b}, \quad \mathbb{A}_{\text{prop},b}, \quad \pi_{\text{prop},b}.$$

WDW-2 constructs all seven. Conversely, a local cotangent reduction alone does not retain source attachment, reconstruction ancestry, boundary fixation, singular objects, transition coherence, or the prohibition on a hidden global selector. The bridge is the proof that the constructed canonical arrows satisfy the imported role predicate. Neither carrier substitutes for the other.

I.2 Exhaustion at the canonical interface

If a proposed descent lacks identities, inverses, compatible composition, associativity, restriction, triple coherence, boundary fixation, quotient stability, or selector exclusion, it fails one of the imported exhaustion classes. If it has all these properties but changes source, reconstruction, charge, or sector data, it fails the canonical preservation predicate. If the change is invisible to the invariant family, it is local presentation skin and is already represented by an arrow. This is the precise sense in which the groupoid is minimal without being declared the final structure of every physical interaction.

I.3 Canonical object and arrow types

A canonical object has the fully typed form

$$\mathbf{O}_U = \left(\begin{array}{l} U, b, \mathcal{P}_b^\dagger|_U, \Omega_b^\dagger|_U, J_{\text{si},b}|_U, \mathcal{C}_{\text{si},b}|_U, \\ \text{Recon}_b|_U, \text{ConnCert}_b, \text{BdyCert}_b, \text{SourceProfile}_b, \text{SectorProfile}_b \end{array} \right).$$

The open or continuation region U is not the physical object by itself. It is the domain on which one local canonical presentation and its witnesses are valid.

An arrow

$$g_{UV} : \mathbf{O}_U \longrightarrow \mathbf{O}_V$$

contains five logically different items:

1. the continuation morphism showing that U and V are comparable;
2. the proper canonical transformation on their common interface;
3. the analytic witness that the transformation acts smoothly tamely on the regular cotangent carrier and respects the Sobolev tower;
4. the Hamiltonian witness that it preserves the spatial/internal zero locus; and
5. the standing witness that reconstruction, source, boundary, and sector profiles agree.

Removing any item changes the arrow type. In particular a continuation morphism without a canonical transformation is only comparability, while a canonical transformation without a standing witness may be a physical symmetry rather than a redundancy.

The corresponding phase object is (U, z) with $z \in U \cap \mathcal{C}_{\text{si}}$. A phase/action arrow is (g_{UV}, z) , not merely g_{UV} . Its source and target are

$$s(g_{UV}, z) = (U, z), \quad t(g_{UV}, z) = (V, g_{UV} \cdot z).$$

This additional coordinate is what types the quotient over the canonical constraint locus.

I.4 Identity

For each $\mathbf{O}_U$, the identity map acts trivially on every canonical coordinate. Its charge and flux vanish, its interface is U , its continuation morphism is the identity, and its preservation witness is reflexivity of every invariant coordinate. Thus

$$1_U : \mathbf{O}_U \rightarrow \mathbf{O}_U$$

is an admitted arrow. This is not a formal embellishment: if the target presentation could not be compared with itself while retaining source and boundary data, it would not have determinate standing in the fixed comparison class.

I.5 Inverse

Let g_{UV} be induced by a proper canonical transformation g . The inverse canonical map g^{-1} is symplectic. Since g lies in the identity component, g^{-1} does also. If

$$Q_{\partial}[g] \equiv 0, \quad \delta Q_{\partial}[g] \equiv 0, \quad \text{Flux}_{\partial}[g] = 0,$$

the inverse carries the same zero profile. Strict equality of source and sector coordinates is symmetric. Reversing the continuation morphism and boundary witness therefore gives

$$g_{VU} = g_{UV}^{-1}.$$

A lossy map or a physical sector-changing transformation fails precisely at this step and is not a proper-redescription arrow.

I.6 Partial composition

Suppose g_{UV} and g_{VW} admit a common restricted interface. The composite canonical map is symplectic and remains in the connected proper transformation family. Constraint preservation is transitive:

$$J(z) = 0 \implies J(g_{UV}z) = 0 \implies J(g_{VW}g_{UV}z) = 0.$$

Structured source, reconstruction, and sector equalities are also transitive. The composite therefore has a preservation witness. If the interfaces have no common restriction, composition is undefined; the groupoid does not invent a global comparison.

I.7 Associativity and neutrality

Canonical maps compose associatively:

$$(g_3 \circ g_2) \circ g_1 = g_3 \circ (g_2 \circ g_1).$$

The two bracketings restrict to the same common interface and carry the same composite preservation witness. Identity maps are neutral. These equalities hold before quotienting skin, so the role-level equalities required by the imported theorem follow a fortiori.

I.8 Restriction

Restriction acts simultaneously on the canonical fields, transformation parameters, continuation interface, reconstruction comparison, boundary witness, and source witness. One is not permitted to restrict the field transformation while retaining an incompatible larger-domain charge certificate. With this synchronized restriction,

$$(g_{VW} \circ g_{UV})|_J = g_{VW}|_J \circ g_{UV}|_J.$$

This equation is the canonical content of restriction compatibility.

I.9 Triple coherence

On a compatible triple interface, compare a direct arrow g_{UW} with $g_{VW} \circ g_{UV}$. If both are admitted as proper arrows between the same objects, then:

coordinate	direct route	composite route
Ω	Ω	Ω
$J = 0$	0	0
Recon	$[\text{Recon}(\rho_W)]$	$[\text{Recon}(\rho_W)]$
Src	$\hat{x}$	$\hat{x}$
Att, W	$(\text{Att}_{\hat{x}}, W_{\hat{x}})$	$(\text{Att}_{\hat{x}}, W_{\hat{x}})$
$Q_{\partial}, \text{Flux}$	$(0, 0)$	$(0, 0)$
Sector	σ	σ

Any remaining representative difference is precisely the presentation skin mediated by the arrow. If a row differs, at least one route is not an arrow of the fixed groupoid. This supplies the imported coherence witness and makes the transition obstruction zero.

I.10 Faithfulness and isotropy

The realization functor need not erase canonical isotropy. An automorphism arrow at an object maps to a role-level automorphism preserving the same invariant profile. Distinct canonical arrows that act identically on every registered invariant may become the same role-level skin arrow, but no difference that changes charge, source, reconstruction, sector, or admitted observable is collapsed. This is the exact quotient-faithfulness required in WDW-2.

I.11 No-selector proof in cover language

Let $\{S_a\}$ be local slices covering the regular orbit-type locus. On $S_a \cap g_{ab}^{-1}S_b$, the transition g_{ab} is an arrow of $\mathbb{G}_{\text{pres}}$, and each point z supplies the action arrow $(g_{ab}, z) \in \mathbb{A}_{\text{prop}}$. On triple overlaps,

$$g_{ac} \sim_{\text{law}} g_{bc} g_{ab}$$

in the boundary-fixed invariant family. A global selector would be a section choosing one representative in every orbit and satisfying additional global compatibility. The proof uses only the cover, local slices, and transition arrows. No such section is constructed or required. Gribov-type failure of a global slice therefore leaves the groupoid theorem intact rather than invalidating it.

I.12 Invariant-map universal property

Let Y be any declared smooth target. A compatible family

$$f_a : U_a \cap \mathcal{C}_{\text{si}} \longrightarrow Y$$

descends through $\mathcal{C}_{\text{si}} // \mathbb{A}_{\text{prop}}$ exactly when

$$f_c(g \cdot z) = f_a(z)$$

for every action arrow (g, z) . Necessity is equality on groupoid-related objects. Sufficiency follows because the equation defines one value on every orbit and is compatible with restriction and composition. On a free regular slice, this is the ordinary quotient submersion universal property. Globally it is retained as a groupoid/stack property, so no Hausdorff orbit space or global representative is presumed.

J. Source-history descent

J.1 History/phase-space type separation

The model phase space contains arbitrary canonical configurations and momenta. The source map is defined only on complete histories solving the model equations and reconstructing smooth covariant fields:

$$\text{Src}_{\mathfrak{M}} : \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) \longrightarrow \hat{Q}_{\text{sm},0}^{\text{src}}.$$

No expression $\mathcal{P}_{\mathfrak{M}} \rightarrow \{\hat{x}\}$ is used. The inverse image

$$\text{Src}_{\mathfrak{M}}^{-1}(\hat{x})$$

is a history fibre and may contain zero, one, or many canonical histories.

J.2 Time-dependent proper presentation change

Let $\varphi_t \in \text{Diff}_0(\Sigma)$ be a smooth path. On spatial fields,

$$h'(t) = (\varphi_t^{-1})^* h(t), \quad \phi'(t) = (\varphi_t^{-1})^* \phi(t),$$

with cotangent-lifted momenta. Lapse and shift transform so that the reconstructed four-metric is the pullback by the foliation-preserving spacetime map

$$(t, x) \mapsto (t, \varphi_t(x)).$$

In particular the shift acquires the velocity of φ_t ; omitting that term would change the reconstructed metric rather than merely its presentation. A history arrow includes this multiplier transformation as a witness coordinate.

J.3 Reconstruction square

For every proper history arrow, the square

$$\begin{array}{ccc} \Gamma & \xrightarrow{g} & g \cdot \Gamma \\ \text{Recon} \downarrow & & \downarrow \text{Recon} \\ (g_{\mu\nu}, \Psi) & \xrightarrow{\tilde{g}} & \tilde{g} \cdot (g_{\mu\nu}, \Psi) \end{array}$$

commutes modulo the WDW-1 foliation-preserving presentation equivalence. Here $\tilde{g}$ denotes the induced spacetime redescription, not an internal gauge element. The Module-A construction is natural under this redescription, so

$$\text{Src}_{\mathfrak{M}}(g \cdot \Gamma) = \text{Src}_{\mathfrak{M}}(\Gamma).$$

The attachment certificate and complete witness are carried as structured coordinates and are not reconstructed from a bare tensor.

J.4 Klein–Gordon pointed fibre

WDW-1 supplies

$$\gamma_\phi \in \text{Src}_\phi^{-1}(\hat{x}_\phi)$$

and a pointed lift

$$\lambda_\phi = (\mathfrak{M}_\phi, \text{Src}_\phi, \gamma_\phi, c_{\lambda_\phi}).$$

The history quotient sends γ_ϕ to $[\gamma_\phi]$. By factorization,

$$\overline{\text{Src}_\phi}([\gamma_\phi]) = \text{Src}_\phi(\gamma_\phi) = \hat{x}_\phi.$$

If another lawful pointed lift has the same source class, it lies in a separate coproduct summand. Proper presentation quotienting does not choose among or identify such lifts.

J.5 Source ancestry through both quotient levels

There are two quotient operations and they must not be conflated. The phase quotient is

$$\mathcal{C}_{\text{si}}^{\text{reg}} \longrightarrow \mathcal{C}_{\text{si}}^{\text{reg}} / \mathcal{G}_{\text{prop}}.$$

It identifies instantaneous canonical presentations. The history quotient is

$$\text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) \longrightarrow \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) // \mathbb{G}_{\text{hist}}.$$

It identifies complete presentation histories whose multipliers and embedding data transform coherently. Evaluation supplies a commutative diagram

$$\begin{array}{ccc} \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) & \xrightarrow{\pi_{\text{hist}}} & \text{Sol}_{\text{can}}^{\text{cl}}(\mathfrak{M}) // \mathbb{G}_{\text{hist}} \\ \text{ev}_t \downarrow & & \downarrow \overline{\text{ev}}_t \\ \mathcal{C}_{\text{si}} & \xrightarrow{\pi_{\text{prop}}} & \mathcal{C}_{\text{si}} // \mathbb{A}_{\text{prop}}. \end{array}$$

The lower arrow alone cannot define the source map because a phase point does not contain the complete on-shell history required by Module A. The upper arrow is the correct source descent domain.

J.6 Attachment and witness retention

The structured source output has the form

$$(\hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}, T_{\hat{x}}).$$

The history action does not reconstruct $\hat{x}$ from the bare transformed tensor. Instead it carries the structured tuple and proves that the reconstructed transformed tensor is a presentation of $T_{\hat{x}}$. Thus the quotient map has the fibrewise form

$$(\Gamma; \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}) \longmapsto ([\Gamma]; \hat{x}, \text{Att}_{\hat{x}}, W_{\hat{x}}).$$

This is the history analogue of the WDW-1 source-fibre-preserving reconstruction theorem.

J.7 Ward ledger after descent

On every complete Klein–Gordon history, Hamilton’s equations imply the scalar field equation, and the metric response satisfies the closed Ward ledger. A proper history arrow pulls the fields back coherently, so the tensor equation and its divergence transform naturally. The descended source class therefore retains the same closed Ward status. This is an implication from the transformed Hamilton equations to the transformed Ward ledger; no converse identification of stress-energy conservation with all matter equations is asserted.

J.8 Multiple histories in one fibre

Nothing in the quotient requires $\text{Src}_{\mathfrak{M}}^{-1}(\hat{x})$ to be a singleton. Distinct complete histories can have the same structured source class while differing in lawful canonical realization data. The history groupoid identifies only proper presentations. If two histories are dynamically or canonically distinct but share a source output, they remain distinct objects over the same source fibre. Source descent is therefore many-to-one where the model permits it and introduces no hidden injectivity claim.

K. Observable descent

Let $\pi : \mathcal{U} \rightarrow T^{\dagger}S_x^{\infty}$ be a regular local quotient.

K.1 Pullback is invariant

For $f \in C^{\infty}(T^{\dagger}S_x^{\infty})$,

$$F = \pi^* f$$

satisfies

$$\rho(\epsilon)F = df(d\pi \rho(\epsilon)) = 0$$

because orbit directions are vertical.

K.2 Invariant functions factor

If F is invariant, it is constant on connected proper orbits. In a slice chart every orbit meets the local slice exactly once on the free locus. Define f by restricting F to the slice and transporting through the quotient chart. Smoothness follows from the slice diffeomorphism, and uniqueness follows from surjectivity of π .

K.3 Poisson compatibility

If $F = \pi^* f$ and $G = \pi^* g$ have Hamiltonian vector fields tangent to the zero locus, then

$$\{F, G\}|_{\mathcal{C}_{\text{si}}} = \pi^* \{f, g\}_{\text{red}}.$$

This follows from $\pi^*\omega_{\text{red}} = \iota^*\Omega$. WDW-2 does not claim that every invariant function on the full singular groupoid has such a global Hamiltonian vector field.

K.4 Separation boundary

Local smooth tame slice charts possess smooth separating coordinates, and each completed Hilbert slice chart possesses smooth functions separating distinct chart points. Across non-Hausdorff global orbit relations, a global smooth separating family may not exist. The exact claim is therefore local invariant/pullback equivalence, not a global maximal observable algebra.

L. WDW-3 handoff schema

Write

$$\mathfrak{A}_b^{\text{reg}} = \{T^\dagger S_a^\infty, \omega_{\text{red},a}^\dagger, T^* S_a^r, \omega_{\text{red},a}^r\}_{a,r \geq r_0}.$$

Table 5: Every WDW-3 handoff coordinate.

Coordinate	WDW-2 status	WDW-3 burden
$\mathcal{C}_{\text{si}}//\mathbb{A}_{\text{prop}}$	CONSTRUCTED	retain groupoid and singular data while adding normal directions
regular local family $\mathfrak{A}_b^{\text{reg}}$	CONSTRUCTED	compare normal evolution across smooth charts and strong local completions
$\overline{\mathcal{O}}_{\text{prop}}$	LOCAL PROVED	identify observables also invariant under the completed normal relation
$\overline{\text{Src}}_{\mathfrak{M}}$	HISTORYWISE PROVED	prove refoliation paths preserve the same structured source class
$\overline{\text{Recon}}$	PROPER-SPATIAL PROVED	extend reconstruction comparison to different embeddings
$\mathcal{H}_\perp[\mathcal{N}]$	EQUIVARIANT FAMILY	compute self-bracket and determine characteristic role
Emb	RETAINED	construct embedding-change groupoid or equivalent deformation object
BdyCert	compact core OCCUPIED	calculate normal boundary generators for every extension
$\{H_\perp[N], H_\perp[M]\}$	OPEN	prove exact closure on the declared field domain
full characteristic equality	OPEN	add normal directions without collapsing standing
refoliation equivalence	OPEN	prove direct/indirect embedding coherence and no selector
clock classification	OPEN DOWNSTREAM	preserve the Physical Clock quotient discipline

Audit condition L.1 (WDW-3 progression rule). WDW-3 may use only branches with WDW-2 status PROVED. It may not replace the groupoid by one slice, treat fixed N as basic, forget source/attachment/boundary indices, or cite the ADM formula as a substitute for a field-theoretic normal-deformation theorem.

L.1 Derivation of normal-family equivariance

On the smooth common domain, the total normal density is a spatial scalar density of weight one. Under ordinary spatial Lie dragging,

$$\delta_\xi \mathcal{H}_\perp = \mathcal{L}_\xi \mathcal{H}_\perp = \xi^i D_i \mathcal{H}_\perp + \mathcal{H}_\perp D_i \xi^i$$

in density notation. Therefore

$$\begin{aligned} \delta_\xi H_\perp[N] &= \int_\Sigma N \mathcal{L}_\xi \mathcal{H}_\perp \\ &= - \int_\Sigma (\mathcal{L}_\xi N) \mathcal{H}_\perp \end{aligned}$$

after integrating the Lie derivative of $N\mathcal{H}_\perp$ over compact Σ . With the Poisson convention fixed in Appendix A, this is $\{H_\perp[N], \mathcal{D}[\xi]\} = -H_\perp[\mathcal{L}_\xi N]$, equivalently the bracket stated in Proposition 33.1. Because the declared left-action fundamental vector is $\rho(\xi) = -X_{\mathcal{D}[\xi]}$, the same calculation gives

$$\rho(\xi)(H_\perp[N]) = H_\perp[\mathcal{L}_\xi N].$$

The important object is the equivariant map

$$\mathcal{N} \longrightarrow C^\infty(\mathcal{D}_b), \quad N \longmapsto H_\perp[N].$$

The spatial group acts on its domain by

$$N \longmapsto (\varphi^{-1})^* N$$

and on its codomain by pullback along the canonical action. Equivariance says that these two actions obey the diagonal identity

$$H_{\perp, \varphi \cdot z}[(\varphi^{-1})^* N] = H_{\perp, z}[N].$$

Differentiating this identity gives the preceding left-action formula and fixes the finite and infinitesimal signs in one convention. It does not say that $H_\perp[N]$ at one fixed arbitrary smearing is constant along the orbit.

L.2 Why the self-bracket is a separate gate

The bracket

$$\{H_\perp[N], H_\perp[M]\}$$

tests relations between two normal deformations. Its standard formal expression involves a spatial generator with metric-dependent smearing. Turning that formula into a fully typed theorem requires:

1. a common field domain stable under both normal Hamiltonian vector fields;
2. the full matter and boundary contributions;
3. control of field-dependent structure functions;
4. a representation on the groupoid-reduced object;
5. an embedding or hypersurface-deformation interpretation; and
6. a source/reconstruction coherence proof for direct and composite refoliation routes.

None follows merely from spatial equivariance. WDW-2 therefore hands the family and these six open coordinates to WDW-3 rather than treating the familiar formal bracket as already closed.

L.3 Branchwise WDW-3 input

For the vacuum branch, the source coordinate is the certified zero-source class. For a Klein–Gordon pointed lift, it is $\overline{\text{Src}}_\phi$ together with the exact fibre and lift index. For a future nontrivial internal branch, the handoff additionally retains the resolved internal quotient, global form, large components, and boundary profile. The three cases are not merged: the WDW-3 theorem must state its normal-deformation result branchwise and may report only the branches whose new certificates are populated.

M. Source-use verification

Every source-use status belongs to

$$\{\text{PROVED, IMPORTED VERIFIED, OCCUPIED TRIVIAL, OCCUPIED NONTRIVIAL,} \\ \text{CONDITIONAL, BLOCKED, OUT OF SCOPE}\}.$$

The following source-use tables state each imported hypothesis, local use, and prohibited stronger reading directly in the publication text.

M.1 Exact import excerpts and prohibited readings

M.1.1 WDW-1 theorem suite

Table 6: WDW-1 import suite.

Theorem	Conclusion used	Prohibited stronger reading
W1.1	accepted Einstein/source dependency and pre-foliation Levi–Civita occupant	does not prove a quotient action
W1.2	unique smooth source projections and Ward ledger from one structured class	does not make source classes phase points
W1.3	normal and tangential Einstein projections	does not prove a moment map
W1.4	gravitational Legendre map, two-form, lapse/shift multiplier status	does not prove a field quotient
W1.5	fibrewise prequotient reconstruction preserving $\widehat{x}$, $\text{Att}_{\widehat{x}}$, $W_{\widehat{x}}$	does not identify distinct spatial presentations
W1.6	pointed strong canonical lifts with model-level phase space and history source map	does not resolve an unproved internal sector
W1.7	compact differentiability and conditional boundary standing	does not calculate every WDW-2 surface charge
W1.8	disjoint handoff and single-valued downstream eligibility	eligibility is not already G_2 closure

M.1.2 Other corpus roles

The Realized Physical Interior supplies the regular coisotropic/characteristic role architecture but not the tame ADM field proof. SM-2 and SYM-1 supply the grammar separating proper internal transformations, boundary-completed Gauss generators, and reduced/Dirac/BRST layers but do not populate a curved-space WDW internal branch. SYM-0 separates constraints, presentation equivalences, physical actions, consequences, and obstructions. SYM-2 requires target updates when spacetime symmetry, tetrads, local Lorentz structure, or boundaries change. The Physical Clock paper prevents a spatial quotient from being reported as the elimination of physical time. The Covariant Differentiation paper preserves connection ancestry but does not define a derivative on superspace. Each role is registered in the source-use tables above.

M.2 Publication acceptance conditions

The endpoint is governed by twenty-two acceptance conditions. Its analytic realization uses the following precise formulation:

Replace “controlled strong Hilbert cotangent completions supporting the spatial reduction” by “a smooth tame/ILH regular-cotangent reduction carrier, with a compatible Sobolev tower and strong Hilbert cotangent completions of every regular local slice quotient.”

This changes no branch, source-family, boundary-standing, quotient, or WDW-3 conclusion.

The endpoint is publication-ready only if all of the following conditions pass:

1. the final WDW-1 publication edition and groupoid source are identified;
2. the forced branch has no route into the WDW-2 input type;
3. every carrier, dual pairing, domain, action, moment map, zero locus, and quotient map is defined, with the declared left action separated from the conventional generator;
4. the spatial generator is derived by variation and $J_{\text{left}} = -\mathcal{D}$ satisfies its signed contraction identity;
5. every claimed nontrivial internal branch has a resolved Dirac ledger;
6. ordinary and gauge-covariant generators are distinguished;
7. the proper generator algebra closes on the common domain;
8. boundary terms and charges are calculated for every occupied branch;
9. large and sector-changing transformations are not silently quotiented;
10. at least one occupied regular chart is populated;
11. the characteristic/orbit/vertical equality is proved using the elliptic tame split, levelwise closed orbit directions, tame zero-locus theorem, direct slice-coordinate kernel calculation, and strong Hilbert completion of each reduced slice chart;
12. the global object is a constructed phase/action groupoid with explicit cover augmentation, action anchor, action, source, target, composition, orbit compatibility, and invariant-map universal property;
13. the canonical bridge instantiates every load-bearing imported clause;
14. no hidden global selector is introduced;
15. source ancestry descends on complete histories;
16. γ_ϕ remains in the exact $\hat{x}_\phi$ fibre;
17. observable separation is not overclaimed beyond regular local charts;
18. the normal family is equivariant with transformed smearing;
19. the Hamiltonian self-bracket and refoliation remain open;
20. abstract, W2.8, conclusion, and ledgers report one endpoint;
21. the PDF contains no unresolved reference, citation, clipped equation, or ledger inconsistency;
22. the publication source reproduces the publication PDF.

Conclusion

WDW-2 closes the proper spatial-redescription gate for the occupied vacuum and attached Klein–Gordon canonical realizations and closes the joint spatial/internal gate for every independently certified internal branch. On each populated free regular chart, proper transformation directions are exactly the characteristic and quotient-vertical directions. The smooth tame regular cotangent carrier makes the action and moment map derivative consistent by keeping the conventional generator $\mathcal{D}$ distinct from the

declared left-action moment map $J_{\text{left}} = -\mathcal{D}$. The elliptic Sobolev tower supplies a compatible family of strong Hilbert completions of the reduced local slices, without asserting completed Hilbert overlap maps. The exact WDW-1 smooth domain remains responsible for curvature and normal-constraint calculations.

The global descent is the explicitly constructed boundary-fixed phase/action groupoid, not a presentation groupoid or presumed smooth orbit manifold. Its cover augmentation, action anchor, action, source, target, composition, orbit, and invariant-map universal properties are typed. Local slices are computational presentations and do not become a global selector. Isotropy and rank-changing objects remain in the groupoid. Boundary charges, large transformations, mapping-class action, and registered sector data are retained. On complete histories, the WDW-1 source-realization family descends unchanged, and the named Klein–Gordon history continues to occupy its exact structured Module-A source fibre.

The resulting object is a source-realization-family-preserving, groupoid-faithful proper-redescription quotient suitable for WDW-3. What passes downstream is the full groupoid and the family of regular local quotient presentations and slice completions together with an equivariant lapse-smeared normal-constraint family, reconstruction and embedding data, source ancestry, boundary status, and a ledger of the still-open normal directions. No normal-deformation closure, refoliation theorem, quantum constraint, anomaly-free quantum representation, physical Hilbert space, or solution to the problem of time is claimed.

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